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Are galaxies in compact groups special?

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ABSTRACT

Context. Compact groups combine high projected densities with low velocity dispersions, but many are embedded in or projected onto larger structures. Their evolutionary significance depends on whether their galaxies remain distinct from regular-group analogues, not only from field galaxies.

Aims. We test whether CG_4 galaxies remain special after removing split systems and comparing them with ordinary four-member groups, luminous regular-group cores, and projected compact cores around brightest group galaxies. We ask which observable carries any residual signal: morphology, star-formation state, colour, AGN-like activity, group-scale structure, phase-space position, or local projected tidal density.

Methods. We use the CG_4 sample inherited from the Tricottet et al. construction, leaving 62 compact groups after the split-system cut, and compare it with RG_4, $\mathrm{Control}_{4B}$, and $\mathrm{Control}_{4C}$. We combine Galaxy Zoo vote-fraction morphologies, SDSS stellar masses and specific star-formation rates, optical colours, emission-line diagnostics, adjusted binomial models, and one-to-one matched controls.

Results. CG_4 galaxies are more often elliptical/smooth and less often spiral/features than control galaxies. This morphology signal survives adjustment for stellar mass, redshift, rank, and available group-scale quantities, and is also present in matched comparisons. The star-formation signal is weaker: CG_4 satellites have a lower raw star-forming fraction, especially relative to RG_4, but this contrast is not comparably robust after adjustment and matching. Among galaxies that remain star-forming, we find no robust compact-group offset from the star-forming main sequence. Colours, AGN-like fractions, magnitude gaps, and the limited Hα diagnostics do not support a simple instantaneous-quenching, fossil-group, or AGN-driven explanation. A projected pairwise tidal-index diagnostic suggests that local tidal density accounts for part of the morphology signal.

Conclusions. Compact groups host a more morphologically evolved population, consistent with accumulated transformation in dense group cores and/or preprocessing. The evidence favours a cumulative structural or morphological signal over a uniquely isolated compact-group quenching mode at the present epoch.

Key words. galaxies: groups: general – galaxies: evolution – galaxies: interactions – galaxies: star formation – galaxies: statistics

1. Introduction

Compact groups of galaxies are useful laboratories for environmental evolution because they combine high projected galaxy densities with relatively low velocity dispersions (Hickson 1982; Hickson et al. 1992). In such configurations, repeated tidal encounters, dynamical friction, mergers, gas stripping or consumption, and disk heating can plausibly accelerate the transformation of galaxies compared with more diffuse environments (Barnes 1985, 1989; Verdes-Montenegro et al. 2001). This expectation is closely connected to the broader morphology-density and star-formation-environment relation, in which galaxy structure and star-formation activity vary systematically with local density (Dressler 1980).

The interpretation of compact groups is nevertheless ambiguous. Projected selection can include chance alignments, and many Hickson-like systems need not be isolated bound entities. Simulations and redshift-space analyses have long shown that apparent compact groups may be transient alignments, dense substructures, or cores embedded in looser groups and richer haloes (Mamon 1986; Díaz-Giménez & Mamon 2010; Díaz-Giménez et al. 2012; Zheng & Shen 2021). The relevant observational question is therefore not only whether compact-group galaxies differ from field galaxies, but whether they remain different from galaxies occupying regular-group analogues selected with comparable luminosity, redshift, and membership constraints.

Recent structural work strengthens this motivation. Using S-PLUS imaging and single-Sérsic measurements, Montaguth et al. (2023, 2025, 2026b,a) identified transition galaxies and trends in the R_e–Sérsic-index plane that point to structural transformation in compact groups, with stronger signatures in non-isolated compact groups and host structures. These studies suggest that compact groups can mark sites of cumulative structural processing. They do not, however, by themselves answer whether compact-group galaxies remain distinct after comparison with regular-group cores selected from the same parent catalogues.

Our previous sample-construction paper (Tricottet et al. 2025) showed that many compact groups are not isolated systems, but appear as dense cores or substructures of richer environments. This result motivates the stricter comparison made here. We use the same compact-group and regular-group sample framework to ask whether CG galaxies are still special after controlling for stellar mass, redshift, galaxy rank, and available group-scale properties. We then ask which observable carries the signal: star-formation class, Galaxy Zoo morphology, optical colour, AGN-like activity, fossil-like magnitude gaps, projected phase-space position, or local projected tidal density.

The paper is organized as follows. Section 2 describes the samples, the sSFR and morphology classifications, and the statistical hierarchy. Sections 3 and 4 present baseline population differences and group-scale diagnostics. Section 5 tests which signals survive adjusted and matched comparisons and examines secondary diagnostics. Section 6 discusses the physical interpretation of the morphology signal, and Section 7 summarizes the conclusions.

2. Data and classifications

2.1. Samples

We use the compact-group and control samples constructed in our previous study Tricottet et al. (2025). We briefly recall their construction here. Galaxy properties are taken from the SDSS DR12 catalogue of Tempel et al. (2017), while regular groups are drawn from the SDSS DR7 group catalogue of Lim et al. (2017); the two catalogues were matched through SDSS object identifiers to define the Lim–Tempel working catalogue. Compact groups are selected from the modified Hickson compact-group catalogue of Díaz-Giménez et al. (2018). Starting from the reliable HMCGs with exactly four members, we retain only systems in which all galaxies are present in the Lim–Tempel catalogue and impose the volume- and luminosity-complete limits $0.005 < z_{\rm BGG} < 0.0452$ and $M_{r,\rm BGG} \leq -21.81$, with all 4 galaxies lying within 3 magnitudes of the brightest group galaxy. This defines the parent $\rm CG_4$ sample of 78 compact groups. The regular-group comparison samples are built from Lim–Tempel groups satisfying the same redshift and BGG-luminosity limits, with at least 4 members and at least 3 satellites within 3 magnitudes of the BGG. From this parent control catalogue, we define $\rm RG_4$ groups as regular groups with exactly four members, excluding systems identical to a $\rm CG_4$, and we additionally use the $\rm Control_{4B}$ and $\rm Control_{4C}$ samples, formed respectively from the 4 brightest galaxies and from the BGG plus its 3 closest projected companions in each parent regular group, after removing groups that share galaxies with $\rm CG_4$s. In contrast to our previous study, we remove CGs classified as *split* following Zheng & Shen (2021). This reduces the compact-group sample from 78 to 62 groups. We also remove the systems associated with Lim group 3688 from our $\rm Control_{4B}$ and $\rm Control_{4C}$ samples. This group contains 6 galaxies with redshifts between 0.032 and 0.034, and 1 with a redshift of 0.048. This group appears to be spurious, and the outlying galaxy, although not the BGG, is selected in both our original $\rm Control_{4B}$ and $\rm Control_{4C}$ samples, significantly affecting the group velocity dispersion (1979 and 1981 $\rm km\,s^{-1}$, respectively).

The three regular-group controls serve different physical purposes. $\rm RG_4$ contains true regular groups with exactly four selected members and is the most direct four-member regular-group comparison. $\rm Control_{4B}$ contains the four brightest galaxies in richer regular groups and tests whether the $\rm CG_4$ population resembles the luminous core of an ordinary group. $\rm Control_{4C}$ contains the BGG plus the three closest projected companions and is the most compact-core-like ordinary-group analogue. $\rm CG_4$ is therefore tested against both equal-membership groups and regular-group cores, rather than against a single control definition.

We use SDSS DR 16 to retrieve stellar masses, star formation rates (SFRs), and Galaxy Zoo morphologies (Lintott et al. 2011). We specifically extract the columns `sfr_tot_p50`, `specsfr_tot_p50`, and `lgm_tot_p50` from the `galSpecExtra` table, and `p_el_debiased` and `p_cs_debiased` from the `zooSpec` table.

2.2. sSFR classification

To build a general star formation classification that could be uniformly applied to all our samples, we extracted galaxies from the SDSS DR 16 using the query shown in Listing 1.

```
SELECT
    s.specObjID,
    s.z,
    p.petroMag_r - p.extinction_r as
        r_obs,
    p.petroMag_u - p.extinction_u as
        u_obs,
    p.petroMag_g - p.extinction_g as
        g_obs,
    p.petroMag_i - p.extinction_i as
        i_obs,
    p.petroMag_z - p.extinction_z as
        z_obs,
    p.objID,
    g.sfr_tot_p50, g.specsfr_tot_p50, g.
        lgm_tot_p50,
    l.h_alpha_eqw, l.h_beta_eqw, l.
        oiii_5007_eqw, l.nii_6584_eqw,
    l.h_alpha_flux, l.h_beta_flux, l.
        oiii_5007_flux, l.nii_6584_flux,
    z.p_el_debiased AS p_E,
    z.p_cs_debiased AS p_S
FROM SpecObj AS s
    JOIN PhotoObj AS p ON s.bestObjID = p
        .objID
    JOIN galSpecExtra as g ON s.specObjID
         = g.specObjID
    JOIN galSpecLine as l ON s.specObjID
        = l.specObjID
    JOIN zooSpec AS z ON s.specObjID = z.
        specObjID
WHERE s.z BETWEEN 0.005 AND 0.0452
    AND (p.petroMag_r - p.extinction_r <=
         17.77)
    AND s.class = 'GALAXY'
    AND g.lgm_tot_p50 > -1000
```

Listing 1. Query used for selecting spectral & photometric data from the SDSS

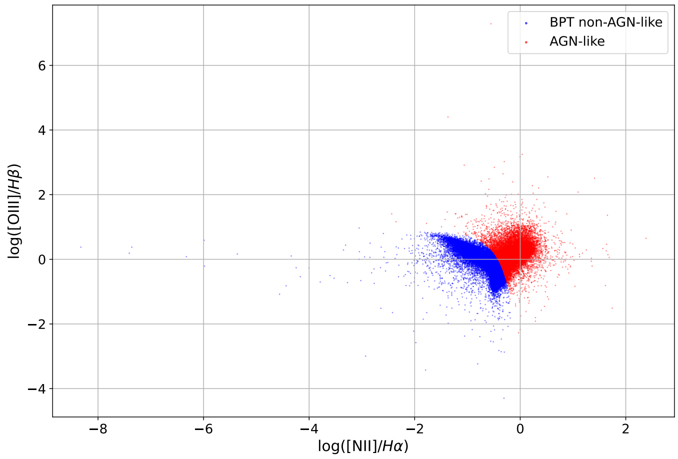

Fig. 1. BPT diagram for galaxies with measured diagnostic line ratios. The AGN-like region is excluded from the sSFR calibration sample; galaxies without the required ratios are BPT-unclassified rather than classified as non-AGN by this diagram.

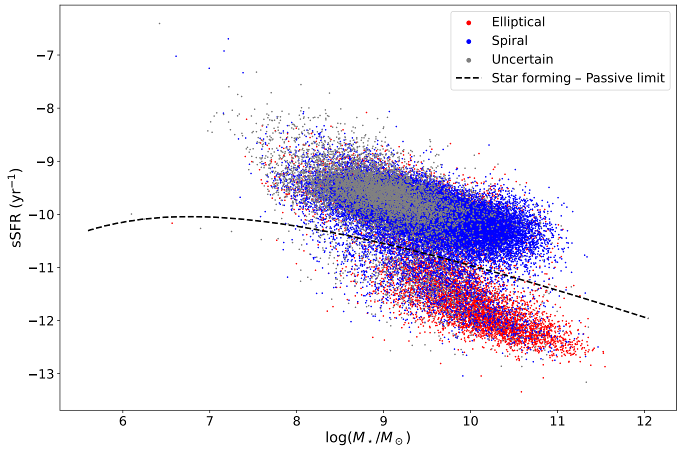

Fig. 2. sSFR vs mass, star-forming/passive limit and Zoo morphologies for the SDSS control sample, excluding quenched galaxies.

We identify BPT-classifiable galaxies by requiring measured values of both emission-line ratios

$$\log_{10}([\mathrm{N\,II}]\lambda6584/\mathrm{H}\alpha) \quad \text{and} \quad \log_{10}([\mathrm{O\,III}]\lambda5007/\mathrm{H}\beta).$$

Among these galaxies, we flag broad AGN-like systems if they satisfy either

$$\log_{10}\frac{[\mathrm{N\,II}]}{\mathrm{H}\alpha} > 0,$$

or if it lies above the empirical demarcation of Kauffmann et al. (2003):

$$\log_{10}\frac{[\mathrm{O\,III}]}{\mathrm{H}\beta} > \frac{0.61}{\log_{10}([\mathrm{N\,II}]/\mathrm{H}\alpha) - 0.05} + 1.3,$$

in which case they are excluded from the GMM calibration sample. Galaxies with missing diagnostic line ratios are BPT-unclassified, not proven non-AGN. They are retained for the broad sSFR status assignment when valid SDSS stellar mass and sSFR estimates are available, while the emission-line/AGN-like analysis below treats BPT-unclassified objects separately. The BPT-classifiable subset is shown in Fig. 1.

Galaxies with `specsfr_tot_p50` equal to -9999 in the SDSS are classified as quenched (for display purposes, they are assigned the arbitrary low value $10^{-14.0}$ Gyr^{-1}). The remaining galaxies are separated into star-forming and passive populations using a two-component Gaussian Mixture Model (GMM) in the $\log M_*$–$\log \mathrm{sSFR}$ plane. Its parameters are determined by minimizing the Kullback–Leibler divergence between the empirical and model densities (Kullback & Leibler (1951); McLachlan & Peel (2000)). For any trial parameter vector $\boldsymbol{\theta}$, we reconstruct the mixture means $\{\boldsymbol{\mu}_i\}$, covariances $\{\boldsymbol{\Sigma}_i\}$, and weights $\{w_i\}$ via a Cholesky-like decomposition and logistic mapping, optionally constraining the second component's mean to isolate the passive population (Dempster et al. (1977)). The KL divergence is estimated by constructing a 2D histogram of the data and summing $p_{\mathrm{data}} \ln(p_{\mathrm{data}}/p_{\mathrm{GMM}})$, with a small regularization ϵ to avoid singularities (Tojeiro et al. (2009); Bisigello et al. (2018)). We optimize $\boldsymbol{\theta}$ via L-BFGS-B (with fallback to Nelder–Mead) over multiple guided and random initializations, selecting the solution with lowest divergence (Pedregosa et al. (2011); Bishop (2006)). Initial means are seeded by a median split in $\log \mathrm{sSFR}$ to improve convergence and robustness against local minima (Rousseeuw (1987)). In the final fit, one component naturally aligns with the high-sSFR "star-forming sequence," while the other captures the low-sSFR "passive" cloud, providing a data-driven division that agrees with previous multi-Gaussian decompositions of the SFR–M_* plane (Wuyts et al. (2011); Hahn et al. (2019)). The decision boundaries, defined as the loci where the posterior probabilities of adjacent components are equal, provide an objective criterion for delineating star-forming from passive galaxies. The GMM calibration excludes galaxies flagged as AGN-like where the required BPT ratios are available, while line-missing galaxies can still receive a broad sSFR class from their SDSS mass and sSFR measurements.

2.3. Raw sSFR fractions

The sSFR status for each sample is shown in Table 1. In the raw all-galaxy comparison, Fisher's exact test gives star-forming-fraction p-values of $p = 6.5 \times 10^{-1}$ versus the Control$_{4\mathrm{B}}$ sample, $p = 2.5 \times 10^{-1}$ versus Control$_{4\mathrm{C}}$, and $p = 1.8 \times 10^{-2}$ versus RG$_4$. We treat these raw tests as descriptive because they do not adjust for stellar mass, rank, or group-scale differences.

The same comparison restricted to BGGs and satellites is also shown in Table 1. For BGGs, the p-values obtained with two-sided Fisher exact test are $p = 0.578$ against Control$_{4\mathrm{B}}$, $p = 0.587$ against Control$_{4\mathrm{C}}$ and $p = 0.37$ against RG$_4$. For satellites, the corresponding p-values are $p = 0.303$ against Control$_{4\mathrm{B}}$, $p = 0.033$ against Control$_{4\mathrm{C}}$ and $p = 2.3\times10^{-4}$ against RG$_4$. The satellite deficit is clearest in the raw comparison with RG$_4$ and is revisited below with adjusted and matched analyses.

2.4. Morphology classification

Morphologies are determined via the Galaxy Zoo citizen-science decision tree, in which each galaxy image receives multiple independent volunteer classifications that are aggregated into debiased vote fractions (Lintott et al.

Sample		Quenched	Passive	Star-forming
CG_4		18 (7.3 %)	146 (58.9 %)	84 (33.9 %)
	BGG	*6 (9.7 %)*	*45 (72.6 %)*	*11 (17.7 %)*
	Satellites	*12 (6.5 %)*	*101 (54.3 %)*	*73 (39.2 %)*
$Control_{4B}$		181 (6.5 %)	1596 (57.1 %)	1019 (36.4 %)
	BGG	*58 (8.3 %)*	*533 (76.4 %)*	*107 (15.3 %)*
	Satellites	*122 (5.8 %)*	*1061 (50.7 %)*	*911 (43.5 %)*
$Control_{4C}$		191 (6.3 %)	1619 (53.8 %)	1198 (39.8 %)
	BGG	*65 (8.7 %)*	*568 (75.6 %)*	*118 (15.7 %)*
	Satellites	*125 (5.5 %)*	*1050 (46.6 %)*	*1078 (47.8 %)*
RG_4		5 (2.2 %)	101 (45.1 %)	118 (52.7 %)
	BGG	*3 (5.4 %)*	*38 (67.9 %)*	*15 (26.8 %)*
	Satellites	*2 (1.2 %)*	*63 (37.5 %)*	*103 (61.3 %)*
SDSS selection		*634 (1.2 %)*	*7705 (14.7 %)*	*44192 (84.1 %)*

Table 1. Number of galaxies in each sSFR status for each sample.

2011). Those fractions are provided in the `zooSpec` table of the SDSS database through the `p_el_debiased` and `p_cs_debiased` columns. We classify a galaxy as "elliptical/smooth" if `p_el_debiased` is greater than 0.5, and as "spiral/features" if `p_cs_debiased` is greater than 0.5. All other cases are classified as "uncertain". These are vote-fraction classes, not a full structural decomposition. In particular, the catalogue used here does not provide a clean E/S0 separation, so references below to elliptical or early-type galaxies should be read as Galaxy Zoo smooth/early-type proxies rather than as a bulge-disc structural classification.

2.5. Statistical strategy

The analyses are organized hierarchically. Raw tables and exact tests provide descriptive baseline comparisons. Fisher or Barnard exact tests are used for two-sample categorical comparisons depending on the table structure, while rank-sum or bootstrap comparisons are used for distributional and median differences. Spearman and Pearson correlations are used only as diagnostics of monotonic or linear trends; when the coefficients are small, we interpret them as weak even if the p-values are formally small. Holm or Benjamini–Hochberg corrections are applied where multiple related tests are performed. The primary robustness layer consists of mass-, rank-, and environment-adjusted binomial models with group-clustered standard errors, followed by one-to-one matched control comparisons. Colours, AGN-like fractions, Hα equivalent widths, magnitude gaps, crossing-time trends, projected phase space, and the pairwise tidal index are secondary diagnostics. The main conclusions below rely on signals that survive the adjusted and matched analyses, not on isolated exploratory tests.

3. Baseline population differences

3.1. Raw morphology fractions

The morphologies obtained for each sample are shown in Table 2. Barnard's two-sided exact tests were used to compare the fraction of each morphology between CG_4 and each control sample. The resulting p-values are $p = 1.4\times10^{-3}$ for ellipticals and $p = 1.2\times10^{-3}$ for spirals versus the $Control_{4B}$ sample, $p = 3.3\times10^{-3}$ and $p = 3.4\times10^{-2}$ versus $Control_{4C}$ and $p = 2.2\times10^{-7}$ and $p = 5.9\times10^{-7}$ versus RG_4. The galaxies in CG_4 systems, which we identified as cores of richer groups in Tricottet et al. (2025), show a robust raw excess of elliptical/smooth classifications and a deficit of spiral/features classifications. Figure 3 displays sSFR versus stellar mass for galaxies in the SDSS selection and in CG_4.

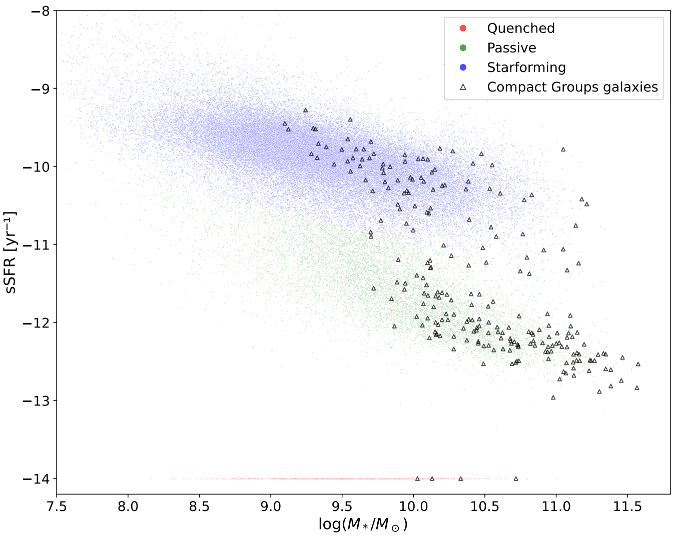

Fig. 3. sSFR and morphologies of galaxies in CG_4 and our SDSS selection. Quenched galaxies are represented at an arbitrary low value of $10^{-14.0}$ Gyr^{-1}.

3.2. Morphology versus sSFR

Among star-forming CG_4 galaxies, 58 are classified as spiral (84.1% after removing galaxies with uncertain morphology) and 11 are classified as elliptical (15.9% after removing galaxies with uncertain morphology). The corresponding numbers are 809 spiral galaxies (88.4%) and 106 elliptical galaxies (11.6%) for $Control_{4B}$, 851 spiral galaxies (84.3%) and 159 elliptical galaxies (15.7%) for $Control_{4C}$, and 91 spiral galaxies (91%) and 9 elliptical galaxies (9%) for RG_4. The probabilities that these fractions are drawn from the same distribution, estimated with Barnard two-sided exact test, are $p = 0.29$ against $Control_{4B}$, $p = 0.97$

Sample	**Elliptical/smooth**	**Spiral/features**	**Uncertain**
CG_4	124 (50.0 %)	86 (34.7 %)	38 (15.3 %)
$Control_{4B}$	1106 (39.6 %)	1266 (45.3 %)	420 (15.0 %)
$Control_{4C}$	1215 (40.4 %)	1249 (41.6 %)	540 (18.0 %)
RG_4	60 (26.8 %)	129 (57.6 %)	35 (15.6 %)
SDSS selection	*10091 (19.2 %)*	*34715 (66.1 %)*	*7725 (14.7 %)*

Table 2. Number of galaxies in each Galaxy Zoo vote-fraction morphology proxy for each sample.

against $Control_{4C}$, and $p = 0.19$ against RG_4. We therefore find neither a significant excess nor a deficit of star-forming elliptical galaxies in CG_4 relative to any control sample.

When restricting the comparison to BGGs, we find 36 elliptical BGGs (58.1% of the BGGs) and 17 spiral BGGs (27.4%) in CG_4. The corresponding numbers are 405 elliptical BGGs (58.0%) and 195 spiral BGGs (27.9%) in $Control_{4B}$, 424 elliptical BGGs (56.5%) and 216 spiral BGGs (28.8%) in $Control_{4C}$, and 21 elliptical BGGs (37.5%) and 26 spiral BGGs (46.4%) in RG_4. The p-values obtained with two-sided Fisher exact test are 1.00 against $Control_{4B}$, 0.88 against $Control_{4C}$ and 0.03 against RG_4. Thus, BGG morphologies differ between CG_4 and RG_4: CG_4 BGGs have a higher elliptical fraction than RG_4 BGGs.

3.3. Population medians by sSFR class

Tables **??**, **??**, and **??** summarize the typical stellar masses, absolute magnitudes, and sSFRs of passive and star-forming galaxies, including splits by the sSFR class of the BGG. These medians are descriptive rather than causal: they show that star-formation class is entangled with stellar mass, luminosity, and galaxy rank, and therefore motivate the adjusted models used below.

3.4. Main sequence

We model the star-forming main sequence as a function of stellar mass using orders 1–4; the polynomial-order comparison is shown in Appendix Fig. A.5. We then use order 2 to compute the residual of each star-forming galaxy with respect to the main sequence, and show their distributions in Fig. 4.

The median main-sequence residual of CG4 galaxies is:

- marginally different from that of Control4B, with a median offset of $\Delta = 0.056$ ($0.029 \leq \Delta \leq 0.099$, bootstrap), corresponding to $p = 0.051$.
- significantly different from that of Control4C, with a median offset of $\Delta = 0.082$ ($0.062 \leq \Delta \leq 0.122$, bootstrap), corresponding to $p = 0.010$.
- not significantly different from that of RG4, with a median offset of $\Delta = 0.046$ ($-0.035 \leq \Delta \leq 0.099$, bootstrap), corresponding to $p = 0.535$.

A direct rank-sum comparison of the star-forming main-sequence offsets only highlights a significant difference between CG_4 and

$Control_{4C}$, whose median offset is -0.060 with respect to the fitted sequence ($p = 0.03$).

Within CG_4, the star-forming main-sequence offsets of the Embedded, Isolated and Predominant classes remain statistically consistent with one another.

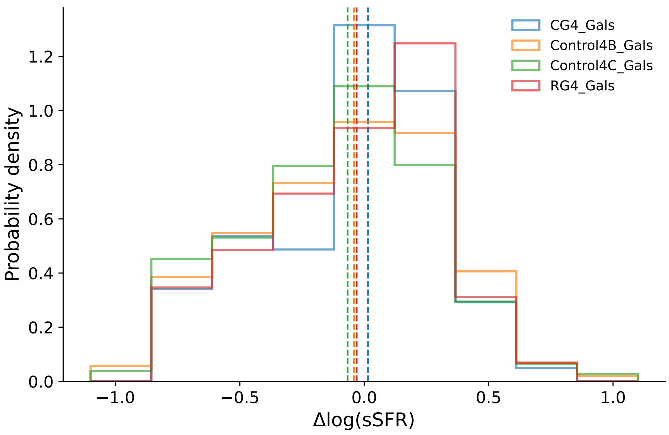

Fig. 4. Main-sequence residual distributions for the star-forming galaxies in each sample.

4. Environmental and group-scale diagnostics

4.1. Distance to the BGG

We also compare the sSFR of satellite galaxies with their distance to the BGG, expressed both in projected kpc and in units of $\langle R_{ij} \rangle$.

Significant monotonic trends are detected in the following cases.

In CG_4, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_S = 0.24$ for 177 satellites ($p = 1.2 \times 10^{-3}$) and Pearson coefficient $r = 0.21$ ($p = 4.1 \times 10^{-3}$).

In $Control_{4B}$, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_S = 0.08$ for 1991 satellites ($p = 3.6 \times 10^{-4}$) and Pearson coefficient $r = 0.08$ ($p = 2.3 \times 10^{-4}$).

In $Control_{4C}$, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_S = 0.25$ for 2143 satellites ($p < 10^{-6}$) and Pearson coefficient $r = 0.20$ ($p < 10^{-6}$).

In $Control_{4C}$, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_S = 0.10$ for 2143 satellites ($p = 4.6 \times 10^{-6}$) and Pearson coefficient $r = 0.09$ ($p = 1.8 \times 10^{-5}$).

The supporting distance plots are shown in Appendix Figs. A.6 and A.7.

4.2. Dominance

We split groups into dominated systems, where the BGG contributes more than 60% of the group luminosity, and non-dominated systems. This split does not produce a strong compact-group signal by itself: CG_4 shows no significant dominated/non-dominated difference after

Sample	$\log_{10}$(sSFR [yr^{-1}])		M_r		$\log(M_\star/M_\odot)$	
	Passive	Star-forming	Passive	Star-forming	Passive	Star-forming
CG4	−12.08 ± 0.03	−10.19 ± 0.07	−21.16 ± 0.14	−20.4 ± 0.14	10.64 ± 0.07	10.05 ± 0.06
Control4B	−12.1 ± 0.01	−10.31 ± 0.02	−21.52 ± 0.04	−20.75 ± 0.04	10.79 ± 0.01	10.24 ± 0.02
Control4C	−11.92 ± 0.02	−10.16 ± 0.02	−20.81 ± 0.07	−19.93 ± 0.04	10.47 ± 0.03	9.82 ± 0.03
RG4	−11.91 ± 0.07	−10.16 ± 0.07	−21.11 ± 0.25	−20.32 ± 0.13	10.62 ± 0.09	9.97 ± 0.09

Table 3. Median values of galaxy properties for passive and star-forming populations in each sample. Uncertainties are estimated by bootstrapping (10 000 iterations).

Sample	$\log_{10}$(sSFR [yr^{-1}])		M_r		$\log(M_\star/M_\odot)$	
BGG	Passive	Star-forming	Passive	Star-forming	Passive	Star-forming
CG4	−11.53 ± 0.09	−10.85 ± 0.29	−20.93 ± 0.11	−20.77 ± 0.32	10.46 ± 0.06	10.32 ± 0.12
Control4B	−11.53 ± 0.03	−10.86 ± 0.06	−21.25 ± 0.03	−21.12 ± 0.07	10.61 ± 0.02	10.46 ± 0.04
Control4C	−11.3 ± 0.03	−10.62 ± 0.06	−20.42 ± 0.05	−20.55 ± 0.08	10.22 ± 0.02	10.15 ± 0.06
RG4	−11.14 ± 0.19	−10.58 ± 0.16	−20.69 ± 0.14	−20.74 ± 0.27	10.32 ± 0.07	10.19 ± 0.13

Table 4. Median values of various quantities for galaxies according to the sSFR class of their BGG (BGG included). Uncertainties are estimated by bootstrapping (10 000 iterations).

Sample	$\log_{10}$(sSFR [yr^{-1}])		M_r		$\log(M_\star/M_\odot)$	
	Passive	Star-forming	Passive	Star-forming	Passive	Star-forming
CG4	−11.23 ± 0.17	−10.92 ± 0.48	−20.45 ± 0.13	−20.34 ± 0.18	10.22 ± 0.05	10.09 ± 0.06
Control4B	−11.26 ± 0.06	−10.91 ± 0.13	−20.9 ± 0.03	−20.75 ± 0.05	10.43 ± 0.02	10.3 ± 0.03
Control4C	−10.96 ± 0.04	−10.6 ± 0.07	−19.8 ± 0.04	−20.02 ± 0.09	9.91 ± 0.02	9.89 ± 0.07
RG4	−10.82 ± 0.19	−10.61 ± 0.22	−20.23 ± 0.16	−20.3 ± 0.13	10.09 ± 0.08	10 ± 0.13

Table 5. Median values of various quantities for galaxies according to the sSFR class of their BGG (satellites only). Uncertainties are estimated by bootstrapping (10 000 iterations).

the Benjamini–Hochberg false-discovery-rate (FDR) correction. Here $p_{\rm adj}$ denotes the FDR-adjusted p-value, and Cliff's δ is a rank-based effect size ranging from -1 to 1.

Across the control and reference samples, the clearest binary separation is instead in group luminosity. Dominated systems have lower $L_{\rm group}$ than their non-dominated counterparts: Control$_{4B}$ gives 9.1×10^{10} versus 1.28×10^{11} $L_\odot$ ($p_{\rm adj} < 10^{-6}$, $\delta = -0.55$); Control$_{4C}$ gives 8.49×10^{10} versus 9.54×10^{10} $L_\odot$ ($p_{\rm adj} < 10^{-6}$, $\delta = -0.25$); and RG$_4$ gives 7.09×10^{10} versus 9.82×10^{10} $L_\odot$ ($p_{\rm adj} = 7.8\times10^{-5}$, $\delta = -0.59$). The negative δ values mean that dominated groups tend to occupy the lower-luminosity side of each comparison.

The spiral-fraction correlations show a broader environmental pattern rather than a clean dominated/non-dominated dichotomy. The most coherent set appears in Control$_{4C}$, where $S_{\rm frac}$ increases with $\langle R_{ij}\rangle$ in both dominated and non-dominated groups ($\rho_{\rm S} = 0.38$, $p < 10^{-6}$ and $\rho_{\rm S} = 0.35$, $p < 10^{-6}$), while weaker negative trends are found with σ_v and $L_{\rm group}$. Control$_{4B}$ contributes only weaker non-dominated-group correlations. In CG$_4$, the significant trends occur in small subsets, with $S_{\rm frac}$ decreasing with $L_{\rm group}$ among dominated groups ($\rho_{\rm S} = -0.50$, 36 groups) and with $\Delta V_{\rm BGG}/\sigma_v$ among non-dominated groups ($\rho_{\rm S} = -0.53$, 26 groups); we therefore treat these as supporting diagnostics rather than the main dominance result. The full distributional and correlation diagnostics are shown in Appendix Figs. A.8 and A.9.

4.3. Morphology and domination

When we further split the samples by domination and by the morphology of the BGG, significant effects only appear in

Control$_{4B}$ dominated groups, where the median satellite spiral fraction is 1.00 for spiral BGGs and 0.67 for elliptical BGGs. The corresponding group-level odds ratio is 1.84 ($p = 0.01$), while the satellite-level clustered fit gives an odds ratio of 1.84 ($p = 0.02$);

Control$_{4C}$ dominated groups, where the median satellite spiral fraction is 0.67 for spiral BGGs and 0.50 for elliptical BGGs. The corresponding group-level odds ratio is 1.84 ($p = 2.1\times10^{-5}$), while the satellite-level clustered fit gives an odds ratio of 1.84 ($p = 2.7\times10^{-5}$).

4.4. Crossing time

The crossing-time diagnostic mainly separates systems by dynamical scale rather than by luminosity alone. Shorter $t_{\rm cross}$ systems have larger virial mass-to-light ratios and virial masses: the correlations with $\log(\mathcal{M}_{\rm VT}/L_r)$ and $\log(\mathcal{M}_{\rm VT})$ are $\rho_{\rm S} = -0.39$ ($p < 10^{-6}$) and $\rho_{\rm S} = -0.33$ ($p < 10^{-6}$), respectively, over 816 groups. By contrast, the relation with total group luminosity is formally significant but very weak ($\rho_{\rm S} = 0.08$, $p = 0.03$). We therefore read $t_{\rm cross}$ as a dynamical compactness indicator tied primarily to virial mass and mass-to-light ratio, with little independent leverage from $L_{\rm group}$. Appendix Figs. A.10–A.12 show the corresponding projected and three-dimensional trends.

4.5. Optical colours

We derive the $(u-r)$, $(u-g)$, $(g-r)$ and $(r-i)$ colours from the matched SDSS photometry. The retained matches include 110 of 248 galaxies in CG_4, 931 of 2792 galaxies in $Control_{4B}$, 1478 of 3004 galaxies in $Control_{4C}$, 102 of 224 galaxies in RG_4. Stellar mass is a confounding variable in colour comparisons (Alpaslan et al. 2015); we therefore model each colour jointly with $\log(M_\star/M_\odot)$ rather than comparing unadjusted colour distributions.

The colour-matched subset is not representative of the full galaxy catalogues: matched galaxies are systematically less massive, more actively star-forming, and more frequently satellites than unmatched galaxies. The fractions with all four colours are 44.4% in CG4, 33.3% in Control4B, 49.2% in Control4C, 45.5% in RG4. The colour analysis is therefore useful as a secondary diagnostic, but it cannot carry the main interpretation by itself.

The raw catalogue colour–mass fits indicate that CG_4 does not simply follow the same colour–mass relation as every comparison sample: common slopes are rejected in 4 of the four colours, with the significant directional slope differences concentrated in the bluer colours and dependent on the adopted comparison catalogue. In the satellite-only comparison, the fitted offsets and slopes likewise depend on the control definition. These tests motivate the robustness analysis rather than establishing a stand-alone compact-group colour effect.

We therefore refit the satellite colours with cluster-robust standard errors, using catalogue-qualified group identifiers. In the pooled comparison adjusted only for stellar mass and redshift, none of the four CG_4 coefficients is significantly negative after Holm correction. After additionally controlling for morphology and sSFR class, negative coefficients appear in $(u-r)$ ($\Delta = -0.127$ mag, Holm-adjusted $p = 0.0177$) and $(u-g)$ ($\Delta = -0.101$ mag, Holm-adjusted $p = 0.0271$). However, these adjusted coefficients are control-sample dependent, and they do not persist within the sSFR or morphology splits. The data therefore do not localize the colour effect to the star-forming spiral population expected for a simple interaction-triggered star-formation scenario.

Residual colour trends with projected BGG-centric distance are consistent with stronger processing near the group centre and a less processed outer population, but these trends use the same biased colour subset and are treated as exploratory. The group-level BGG–satellite colour concordance is similarly secondary: redder BGGs tend to have redder mean satellite populations in the combined group sample, but the CG_4 slope is not a robustly distinct population-level signature. Detailed colour–mass, satellite, radial, BGG–satellite, and domination diagnostics are shown in Appendix Figs. A.13–A.17. Figure 5 summarizes the model and control-sample dependence that motivates our interpretation: the present data do not establish a robust global blue or red offset for CG_4 satellites.

5. Robustness: what drives the compact-group differences?

The previous sections establish that CG_4 galaxies differ from the control catalogues in their population mix, especially in morphology and in the star-forming fraction of satellites. We now ask whether these differences survive after controlling for stellar mass, rank and available group-scale quantities, and whether they are better interpreted as compact-group-specific evolution, preprocessing in dense halo cores, or catalogue-selection effects.

5.1. Mass-, rank- and environment-adjusted galaxy properties

We fit binomial models to the combined galaxy catalogues, standardizing the continuous predictors and using group-clustered standard errors. The common adjustment set includes stellar mass, redshift, satellite/BGG rank, group luminosity and velocity dispersion when sufficiently complete.

For passive status, the CG_4 odds ratio is 1.20, with 95% confidence interval $[0.88, 1.63]$ and $p_{\mathrm{Holm}} = 0.485$.

For passive satellite status, the CG_4 odds ratio is 1.31, with 95% confidence interval $[0.95, 1.80]$ and $p_{\mathrm{Holm}} = 0.390$.

For star-forming satellite status, the CG_4 odds ratio is 0.76, with 95% confidence interval $[0.56, 1.05]$ and $p_{\mathrm{Holm}} = 0.390$.

For elliptical morphology, the CG_4 odds ratio is 1.62, with 95% confidence interval $[1.21, 2.16]$ and $p_{\mathrm{Holm}} = 0.007$.

For elliptical BGG morphology, the CG_4 odds ratio is 0.90, with 95% confidence interval $[0.46, 1.74]$ and $p_{\mathrm{Holm}} = 0.751$.

For spiral morphology, the CG_4 odds ratio is 0.62, with 95% confidence interval $[0.46, 0.83]$ and $p_{\mathrm{Holm}} = 0.007$.

Compact-group membership therefore remains independently associated with morphology after the available adjustments: CG_4 galaxies have a higher probability of being elliptical and a lower probability of being spiral. In contrast, the adjusted passive and star-forming satellite coefficients are weaker and do not provide evidence for an instantaneous star-formation difference that remains significant after correction.

5.2. Matched-control comparison

We matched 234 CG_4 galaxies one-to-one to 234 ordinary-group galaxies without replacement, requiring common propensity-score support, a 0.2-standard-deviation caliper and exact rank strata. The propensity model uses logMstar, z numeric, rank, log group luminosity, velocity dispersion. Projected distance is retained as a phase-space and interaction diagnostic rather than matched away because it partly defines compactness and has limited overlap. The largest absolute standardized mean difference changed from 0.28 before matching to 0.14 afterwards.

The matched CG_4–minus–control difference in passive fraction is 0.098, with 95% bootstrap interval $[0.017, 0.175]$ and $p_{\mathrm{Holm}} = 0.054$.

This is suggestive but not formally significant after correction, so we treat the passive-fraction excess as borderline.

The corresponding CG_4–minus–control difference in star-forming fraction is -0.098, with 95% bootstrap interval $[-0.175, -0.017]$ and $p_{\mathrm{Holm}} = 0.054$.

The matched CG_4–minus–control difference in elliptical fraction is 0.197, with 95% bootstrap interval $[0.090, 0.303]$ and $p_{\mathrm{Holm}} < 10^{-6}$.

The matched CG_4–minus–control difference in spiral fraction is -0.197, with 95% bootstrap interval $[-0.303, -0.090]$ and $p_{\mathrm{Holm}} < 10^{-6}$.

Among matched star-forming galaxies, the residual-sSFR difference is -0.096, with 95% bootstrap interval $[-0.221, 0.015]$ and $p = 0.084$.

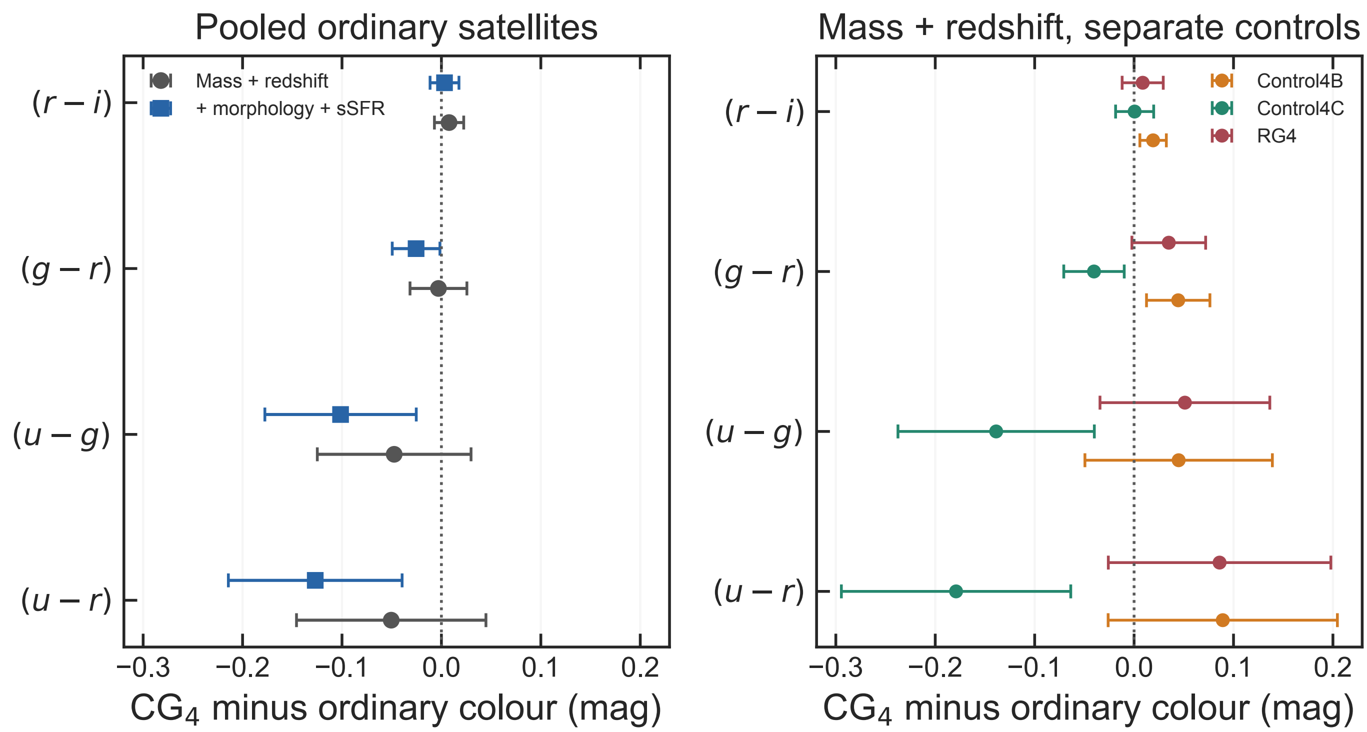

Fig. 5. Robustness of the satellite colour coefficients. Left: pooled ordinary-group comparison before and after controlling for morphology and sSFR class. Right: mass- and redshift-adjusted comparisons against each control catalogue separately. Points show the CG_4 minus ordinary coefficient and bars show 95% confidence intervals; negative values indicate bluer CG_4 satellites.

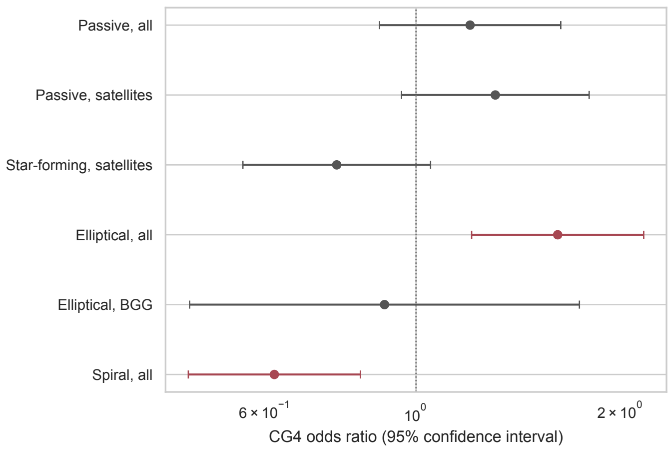

Fig. 6. Adjusted CG_4 odds ratios. Bars show 95% confidence intervals; the vertical line marks equal odds.

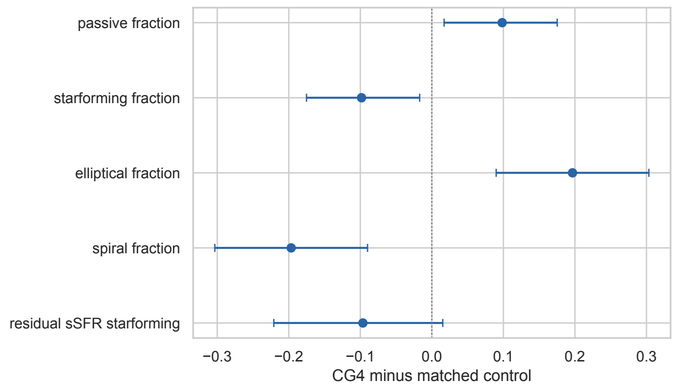

Fig. 7. CG_4–minus–control effects after one-to-one matching. Bars show 95% bootstrap confidence intervals.

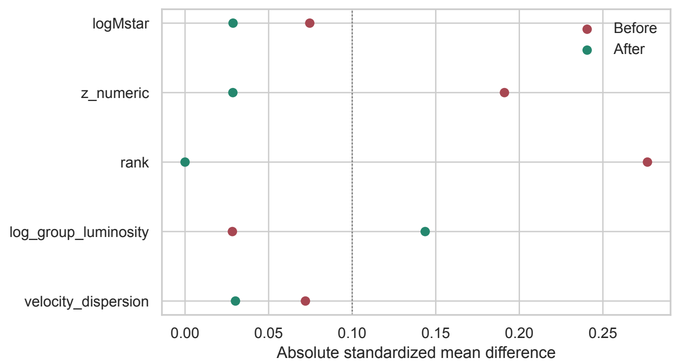

Fig. 8. Absolute standardized mean differences before and after matching.

The main interpretation does not rely on the exact GMM division of the sSFR plane. Repeating the key satellite comparison with a fixed threshold, $\log_{10}(\mathrm{sSFR/yr^{-1}}) = -11.0$, gives CG_4 and RG_4 satellite star-forming fractions of 40.7% and 60.2%, respectively, a CG_4–RG_4 difference of -19.6 percentage points ($p = 3.6 \times 10^{-4}$). The qualitative hierarchy is unchanged: the morphology result is independent of the sSFR threshold, whereas the passive/star-forming contrast remains more sensitive to modelling choices.

The matched comparison is therefore consistent with the adjusted models: the most robust compact-group signal is morphological, while the passive and star-forming-fraction signals are more model-dependent.

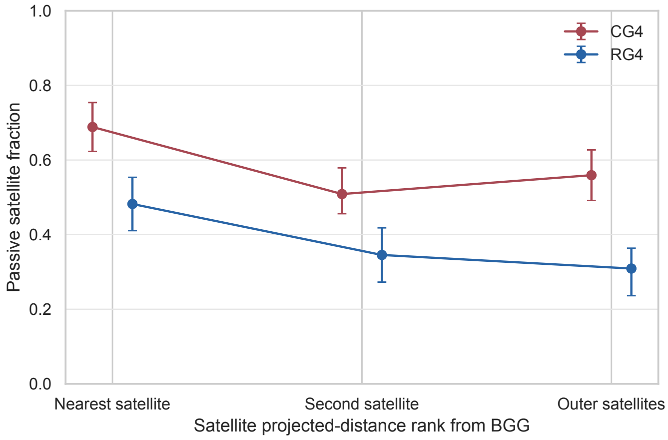

Fig. 9. Passive satellite fraction as a function of projected-distance rank from the BGG. Points show CG_4 and RG_4 fractions; bars show group-cluster bootstrap 16th–84th percentile intervals.

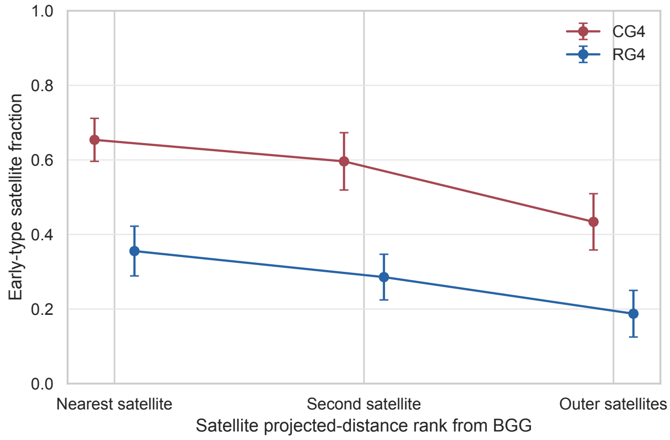

Fig. 10. Early-type satellite fraction as a function of projected-distance rank from the BGG. Because the catalogue separates ellipticals, spirals, and uncertain morphologies but not S0 galaxies, early type is equivalent to elliptical in this diagnostic.

5.3. Satellite segregation around the brightest group galaxy

The previous sections show global differences between CG_4 and RG_4 satellites. We now ask whether these differences are concentrated near the BGG, as expected for accelerated evolution in the dense inner group, or whether they persist across the satellite population. The analysis uses 177 CG_4 and 166 RG_4 satellites, excludes rank-1 galaxies, ranks satellites by projected BGG-centric distance, and measures passive and early-type fractions with group-cluster bootstrap uncertainties. When available, we also use $|\Delta v_{\rm BGG}|/\sigma_v$, with $\Delta v_{\rm BGG} = c(z_{\rm gal} - z_{\rm BGG})/(1 + z_{\rm group})$, as a projected phase-space diagnostic. Similar projected phase-space coordinates are often used as observational proxies for satellite infall state or quenching time, but only statistically (Mahajan et al. 2011; Oman & Hudson 2016; Wetzel et al. 2013).

In the innermost satellite-distance bin, the passive fractions are 68.9% for CG_4 and 48.2% for RG_4, a CG_4–RG_4 difference of 20.6 percentage points. The corresponding early-type fractions are 65.4% and 35.6%, with a difference of 29.8 percentage points.

The passive-fraction contrast is similar between the inner and outer distance ranks, so the data do not isolate the CG_4 offset to the immediate BGG environment.

After adjusting for stellar mass, redshift, distance bin, and available group-scale controls, the passive-status model gives a CG_4 odds ratio of 1.27 (95% CI 0.47–3.46; p = 0.640).

The analogous early-type model gives a CG_4 odds ratio of 2.57 ($p = 0.058$).

A nearest-neighbour mass–redshift matched check uses 128 CG_4–RG_4 satellite pairs. The matched passive-fraction difference is 9.4 percentage points, and the matched early-type difference is 19.6 percentage points.

Projected phase-space coordinates should not be read as orbital reconstructions for individual galaxies. In compact groups the member counts are small, velocity dispersions are noisy, and projection mixes recently accreted and long-resident satellites. We therefore treat these bins as an observational diagnostic of where the CG_4–RG_4 population contrast appears, rather than as a direct dynamical clock.

5.4. Magnitude gaps and fossil-like assembly

We define $\Delta m_{12} = M_{r,2} - M_{r,1}$ and $\Delta m_{14} = M_{r,4} - M_{r,1}$, so that a more dominant BGG has a positive, larger gap.

For Δm_{12}, the CG_4 and control medians are 1.17 and 1.09 mag, respectively ($p_{\rm Holm} = 0.705$).

For Δm_{14}, the CG_4 and control medians are 2.56 and 2.59 mag, respectively ($p_{\rm Holm} = 0.705$).

The compact-group morphology signal is therefore not accompanied by a significant excess in these simple fossil-like magnitude-gap indicators.

The correlation between Δm_{12} and passive satellite fraction is $\rho_S = -0.15$ ($p_{\rm Holm} < 10^{-6}$).

The supporting magnitude-gap distributions are shown in Appendix Figs. A.19 and A.20.

5.5. Recently quenched candidates

The catalogues do not contain both D_n4000 and $H\delta_A$, so a post-starburst classification is not possible. We therefore perform only a limited $H\alpha$-equivalent-width comparison, using the SDSS convention in which negative equivalent width denotes emission. We classify strong $H\alpha$ emission using h_alpha_eqw <= -3 Angstrom.

The strong-emission fraction is 32.3% in CG_4, 41.0% in Control4B, 45.7% in Control4C, 56.9% in RG4.

Relative to Control4B, the CG_4 strong-emission fraction differs by −0.087 ($p_{\rm Holm} = 0.012$).

Relative to Control4C, the CG_4 strong-emission fraction differs by −0.135 ($p_{\rm Holm} = 2.4 \times 10^{-4}$).

Relative to RG4, the CG_4 strong-emission fraction differs by −0.246 ($p_{\rm Holm} = 1.5 \times 10^{-6}$).

These limited $H\alpha$ results indicate that CG_4 galaxies have a lower strong-emission fraction than in each control sample in the current catalogue, but they do not provide a direct post-starburst or recently-quenched classification.

The $H\alpha$ equivalent-width distributions are shown in Appendix Fig. A.21.

5.6. Emission-line and AGN-like populations

The complementary emission-line analysis classifies BPT-star-forming, composite and broad AGN-like populations for galaxies with the required diagnostic line ratios. BPT-unclassified galaxies are excluded from the AGN-like fraction denominator rather than treated as confirmed non-AGN. Among BPT-classified CG_4 galaxies, the AGN-like fraction is 38.0%.

After mass and environment adjustment, the CG_4 AGN-like odds ratio is 1.14 ($p = 0.451$).

Thus, AGN-like activity does not provide a robust independent compact-group discriminator in this analysis.

The AGN-like fractions are shown in Appendix Fig. A.22.

5.7. Pairwise tidal index

The pairwise analysis reconstructs projected separations and velocity differences for 6268 galaxies and tests whether adding $\sum_j M_{\star,j}/R_{ij}^3$ changes the adjusted CG_4 coefficient.

For passive status, the baseline CG_4 odds ratio is 1.20 ($p = 0.227$), while the model including the tidal index gives a CG_4 odds ratio of 0.80 ($p = 0.170$). The tidal-index term itself has odds ratio 1.65 ($p < 10^{-6}$).

For elliptical morphology, the baseline CG_4 odds ratio is 1.65 ($p = 4.1 \times 10^{-4}$), whereas the model including the tidal index gives a CG_4 odds ratio of 1.15 ($p = 0.346$). The tidal-index term has odds ratio 1.56 ($p < 10^{-6}$). This suggests that the reconstructed local tidal-density proxy absorbs much of the unadjusted CG_4 morphology coefficient, although the result remains an exploratory projected-pair diagnostic.

5.8. Selection diagnostics

The fractions with complete measurements for the major derived quantities are shown in Appendix Fig. A.23. The colour-matched versus unmatched comparisons quantify shifts in stellar mass, redshift, rank, star-formation class and morphology.

The median nearest-neighbour angular separation is 100.9 arcsec in CG_4 and 209.0 arcsec in the controls; the fractions below the 55-arcsec SDSS fibre-collision scale are 19.4% and 9.3%, respectively.

No major availability difference meets both criteria of corrected significance and a difference exceeding 10 percentage points.

The group-scale availability row is based on the group covariates available for this catalogue: radius scale, velocity dispersion, group luminosity, and BGG luminosity dominance. A harmonized group-mass estimate is not available in these tables, so models that request it omit that covariate.

The colour-selection mass audit is shown in Appendix Fig. A.24.

6. Discussion: drivers of structural evolution in compact groups

6.1. Local projected tidal density and morphology signal

The most robust residual difference between CG_4 and the regular-group controls is morphological. CG_4 galaxies are more likely to be classified as elliptical/smooth and less likely to be classified as spiral/features, and this result persists in both the adjusted binomial models and the one-to-one matched comparison. The result hierarchy therefore points to accumulated structural transformation more strongly than to a present-day star-formation offset.

The pairwise tidal-index experiment strengthens this interpretation while also narrowing it. For elliptical/smooth morphology, the baseline CG_4 odds ratio is 1.65 ($p = 4.1 \times 10^{-4}$). After adding the projected pairwise tidal-index proxy, the CG_4 odds ratio decreases to 1.15 ($p = 0.346$), while the tidal-index term itself has odds ratio 1.56 ($p < 10^{-6}$). This behaviour is expected if compact groups are selecting galaxies in locally dense projected configurations where repeated close companions, rather than the catalogue label alone, are linked to the smooth/early-type-like population. Because the proxy is reconstructed from projected separations and catalogue masses, it should be read as a local projected tidal-density diagnostic, not as a direct measurement of the full time-dependent tidal field.

6.2. Smooth morphologies, S0 degeneracy, and transition galaxies

The Galaxy Zoo morphology variables used here separate smooth, featureless systems from galaxies with visible spiral structure or other features. They do not cleanly distinguish classical ellipticals from S0 galaxies, faded disks, or other smooth disk-dominated systems. The enhanced CG_4 elliptical/smooth fraction should therefore be interpreted as a shift toward smooth or early-type-like appearances, not as proof that compact groups specifically build classical elliptical galaxies.

This caveat is important when comparing with the recent S-PLUS structural studies of compact-group galaxies by Montaguth et al. (2023, 2025, 2026b,a). Those studies identify transition galaxies and trends in the R_e–Sérsic-index and mass-size planes, with especially strong signals in non-isolated compact groups and host structures. Their transition population can plausibly overlap with the smooth class in our catalogue: systems with faded disks, increasing central concentration, or S0-like structure may be counted as smooth here even if they are not classical ellipticals. The two approaches are therefore complementary. Their structural measurements describe the likely physical form of the transformation, while our regular-group controls show that the strongest residual compact-group signal is the accumulated smooth/featured morphology balance.

6.3. Decoupling between morphology and instantaneous star formation

The star-formation diagnostics do not follow the morphology signal one-to-one. CG_4 satellites have a lower raw star-forming fraction, especially relative to RG_4, but the adjusted and matched evidence is weaker than for morphology. Among galaxies that remain star-forming, the main-sequence residuals do not show a robust compact-group offset. The colour analysis is also control-sensitive and based on a biased colour-matched subset, while AGN-like fractions, magnitude gaps, and the limited Hα diagnostic do not identify a simple alternative driver.

This decoupling suggests that the dominant observable difference is not an instantaneous quenching event caught at the present epoch. A galaxy can lose visible spiral structure, build a smoother disk or bulge-dominated profile, or

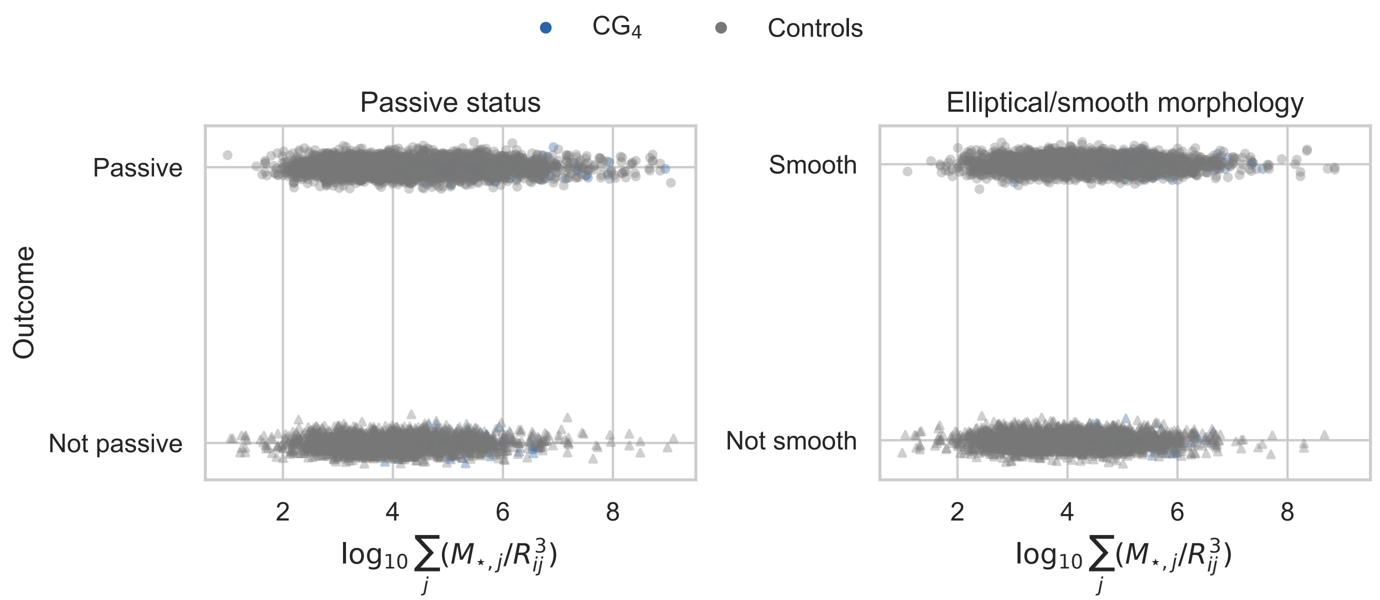

Fig. 11. Passive status and elliptical/smooth morphology as functions of the reconstructed pairwise tidal index. Colours distinguish CG$_4$ from control galaxies.

Diagnostic	Baseline CG$_4$ signal	Survives adjusted/-matched analysis?	Interpretation
Elliptical/smooth fraction	Enhanced in CG$_4$	Yes	Robust accumulated morphology signal.
Spiral/features fraction	Reduced in CG$_4$	Yes	Reduced late-type vote-fraction population.
Satellite star-forming fraction	Reduced, especially versus RG$_4$	Weak or borderline	Partly explained by mass, rank, environment, or preprocessing.
Star-forming main-sequence residual	Weak and control-dependent	No robust offset	No strong instantaneous SFR enhancement or suppression among remaining star-forming galaxies.
Satellite colours	Colour-slope and residual hints	Control-sensitive	Exploratory; not a simple uniform reddening or blueing of CG$_4$ satellites.
AGN-like fraction	No robust independent signal	No	Not an AGN-driven explanation.
Magnitude gaps	No significant excess	No	Not fossil-like under the simple gap diagnostics used here.
Pairwise tidal index	Local projected density proxy is informative	Absorbs much of the morphology coefficient	Local tidal density likely contributes to the morphology signal.

Table 6. Result hierarchy. Baseline tests describe where CG$_4$ differs from the controls; the main interpretation is based on diagnostics that remain after adjusted or matched comparisons.

fade into an S0-like state on timescales that do not require a simultaneously large sSFR residual in the surviving star-forming subset. The present data therefore favour a cumulative morphology or structure signal over a single ongoing star-formation process.

6.4. Compact groups as dense cores and preprocessing sites

The comparison samples were designed to ask whether compact groups remain special relative to ordinary group cores, not merely relative to the field. The answer is mixed but physically useful. CG$_4$ galaxies retain a robust morphology difference, yet much of the interpretation points toward local density, dense cores, and preprocessing rather than toward an isolated compact-group channel. This is consistent with the construction result that many compact groups are embedded in or associated with larger structures, and with the stronger structural signatures reported for non-isolated compact groups in recent S-PLUS work.

The most conservative interpretation is that compact groups identify the high-density, high-interaction tail of group environments. Some systems may be genuinely compact and dynamically evolved, while others may be dense cores or substructures inside larger haloes. In both cases, the relevant physical quantity is likely the accumulated local interaction history. Future simulations and direct structural measurements, including R_e, Sérsic index, transition-galaxy classifications, and mass-size residuals, are needed to test whether the signal isolated here is unique to compact configurations or is the high-density limit of more general group preprocessing.

7. Conclusion

We have tested whether galaxies in CG_4 compact groups remain special after removing split systems and comparing them with three regular-group analogues: RG_4, $Control_{4B}$, and $Control_{4C}$. The comparison was designed to separate compact configurations from ordinary four-member groups, luminous group cores, and projected compact cores around BGGs.

The main result is morphological. CG_4 galaxies have an enhanced elliptical/smooth Galaxy Zoo fraction and a reduced spiral/features fraction. This signal survives the mass-, redshift-, rank-, and group-adjusted binomial models and is also recovered in the one-to-one matched-control analysis.

The star-formation result is weaker. CG_4 satellites have a lower raw star-forming fraction, especially relative to RG_4, but the adjusted and matched evidence is weaker or borderline. Among galaxies that remain star-forming, CG_4 galaxies do not show a robust offset from the fitted star-forming main sequence.

The secondary diagnostics do not identify a simple alternative driver. The colour results are control-sensitive and based on a biased colour-matched subset. AGN-like fractions and magnitude gaps do not explain the morphology signal, the Hα diagnostic is limited by missing age-sensitive spectral indices, and projected phase-space trends do not localize the passive difference uniquely to the nearest satellites.

We therefore interpret CG_4 galaxies as a more morphologically evolved population, consistent with accumulated transformation in very dense environments, dense group cores, and/or preprocessing. The pairwise tidal-index experiment suggests that local projected tidal density accounts for part of the compact-group morphology coefficient, so the result should not be read as evidence for a uniquely isolated compact-group quenching mechanism.

Future work should connect this regular-group control framework directly to structural measurements, including R_e, Sérsic index, transition-galaxy classifications, mass-size residuals, and better spectroscopic post-starburst diagnostics. Those measurements would test whether the morphology signal isolated here is the same cumulative structural transformation highlighted by recent S-PLUS studies.

Appendix A: Supplementary diagnostic figures

This appendix collects the secondary diagnostic figures referenced in the main text.

A.1. sSFR and main-sequence diagnostics

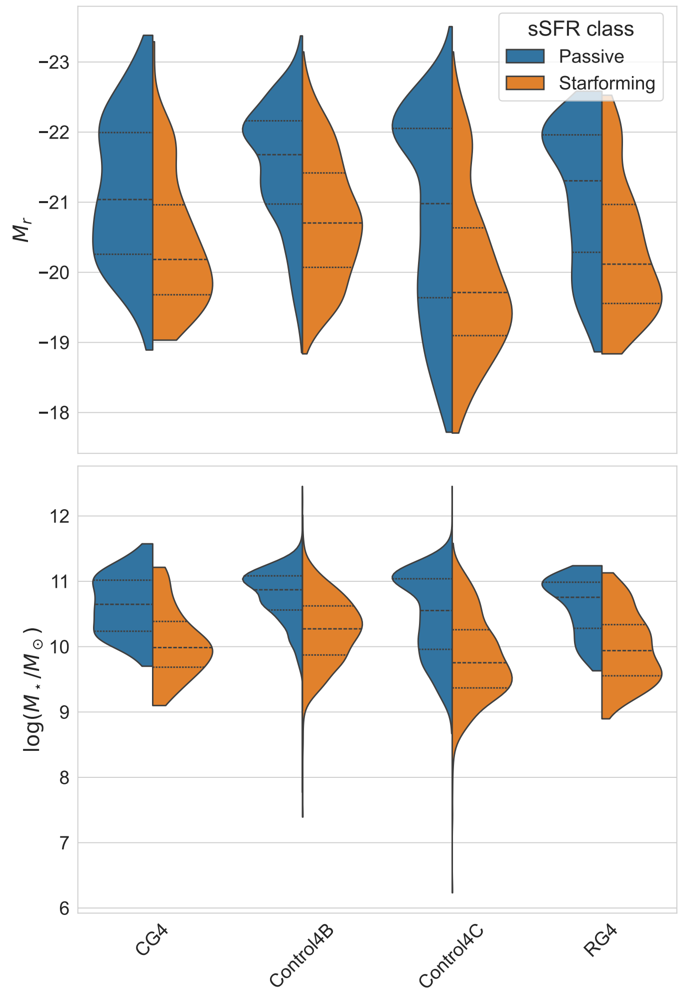

Fig. A.2. Distributions of various quantities for galaxies according to their sSFR class.

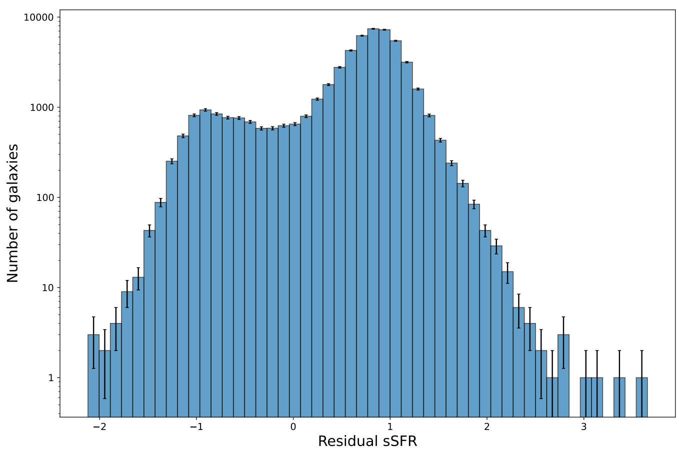

Fig. A.1. Distribution of residual sSFR offsets from the adopted star-forming main sequence.

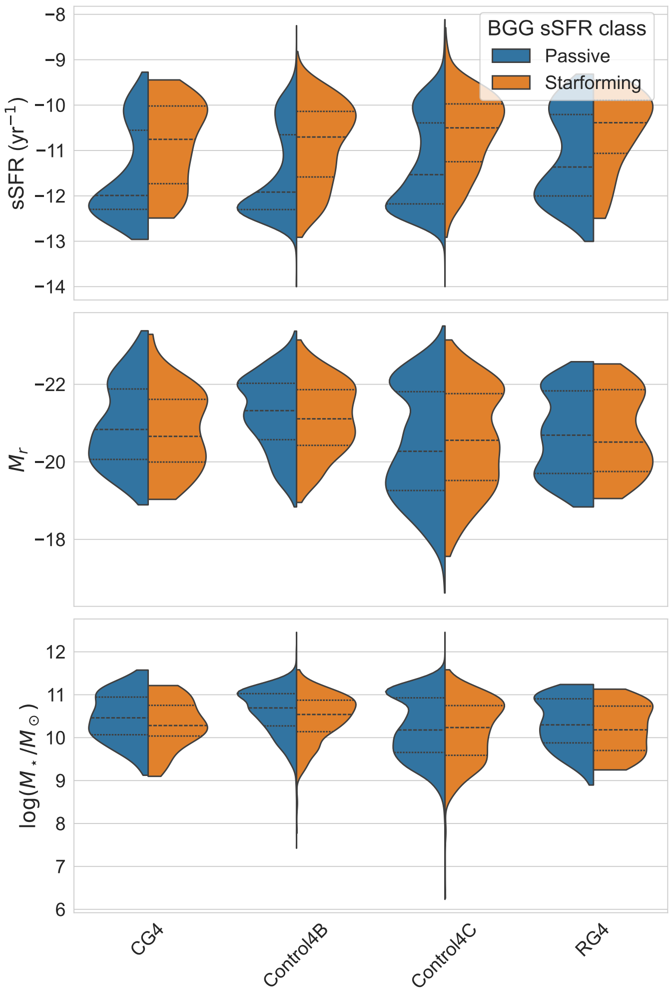

Fig. A.3. Distributions of various quantities for galaxies according to the BGG sSFR class, with BGGs included.

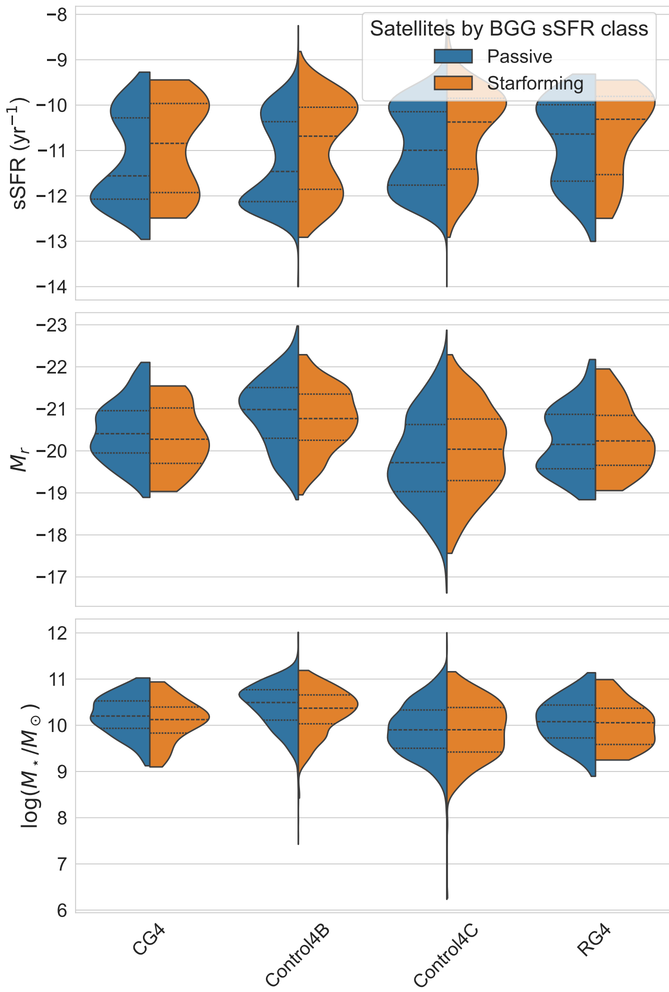

Fig. A.4. Distributions of various quantities for satellites according to the BGG sSFR class.

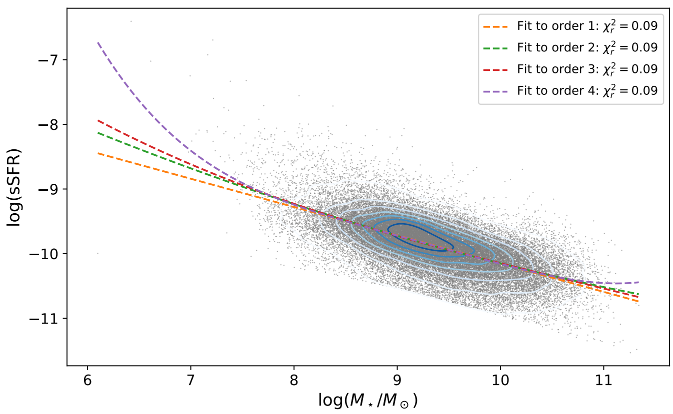

Fig. A.5. Polynomial fits of orders 1–4 to the star-forming main sequence.

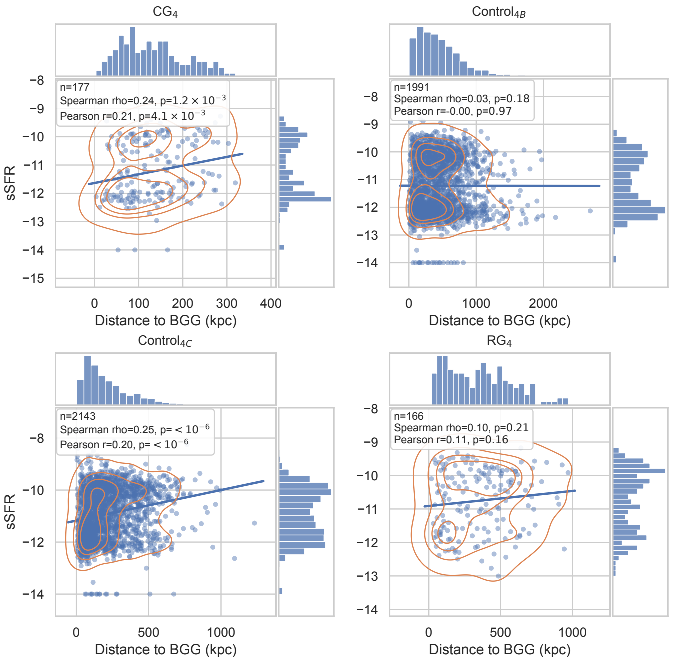

Fig. A.6. sSFR as a function of projected distance to the BGG, in kpc, for satellite galaxies in each sample.

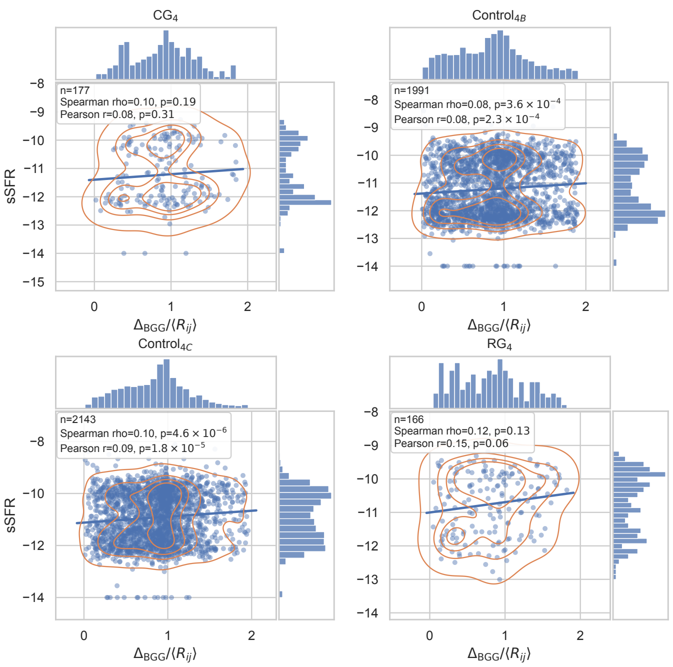

Fig. A.7. sSFR as a function of projected distance to the BGG normalized by $\langle R_{ij} \rangle$ for satellite galaxies in each sample.

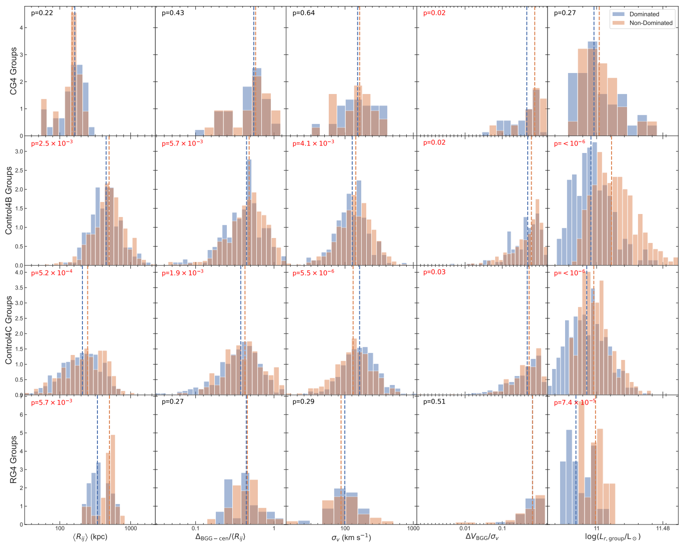

Fig. A.8. Distributions of the main group-scale observables in dominated and non-dominated systems.

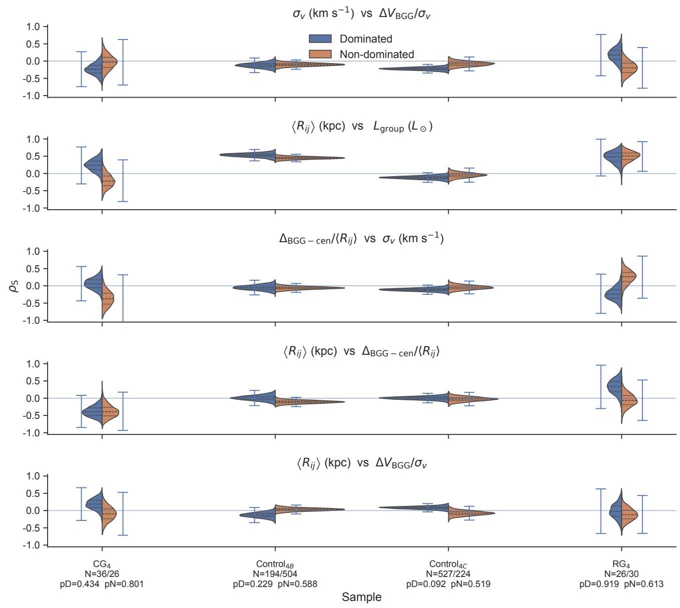

Fig. A.9. Bootstrap distributions of the Spearman coefficients for the group-property pairs selected by the XOR domination criterion.

A.2. Group-scale and radial diagnostics

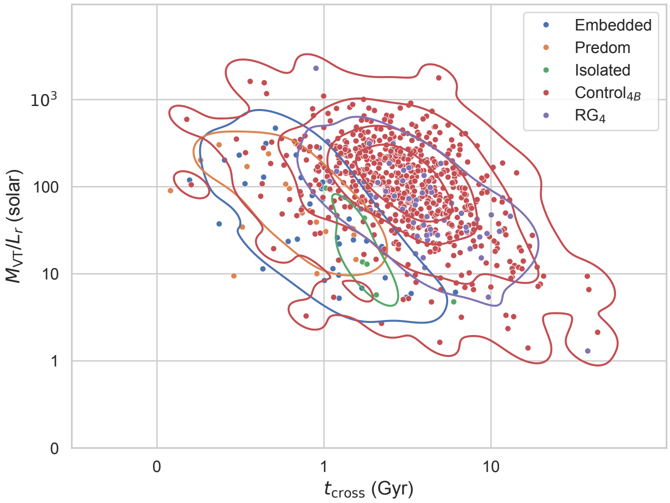

Fig. A.10. Crossing time versus virial mass-to-light ratio for the compact groups and their controls.

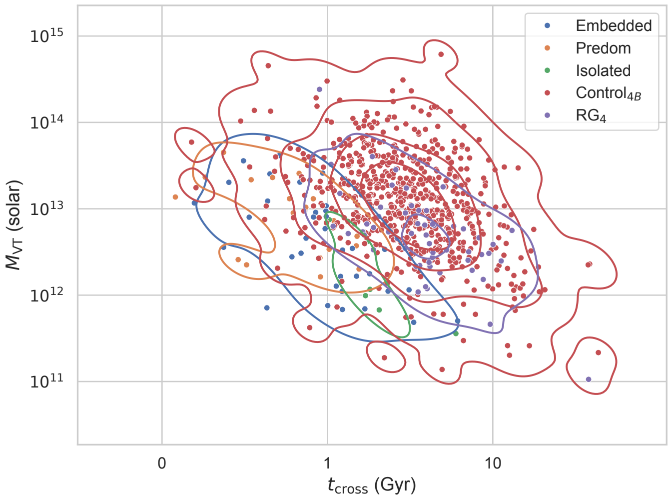

Fig. A.11. Crossing time versus virial mass for the compact groups and their controls.

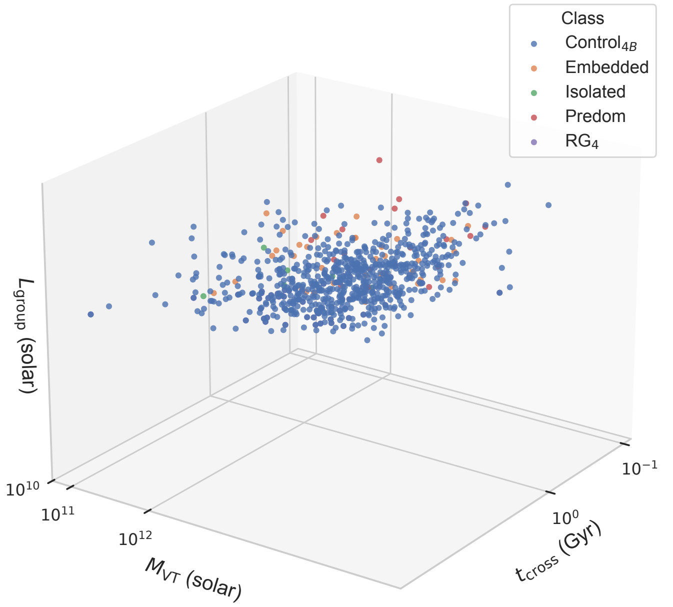

Fig. A.12. Three-dimensional view of the crossing time, virial mass, and group luminosity space.

A.3. Colour diagnostics

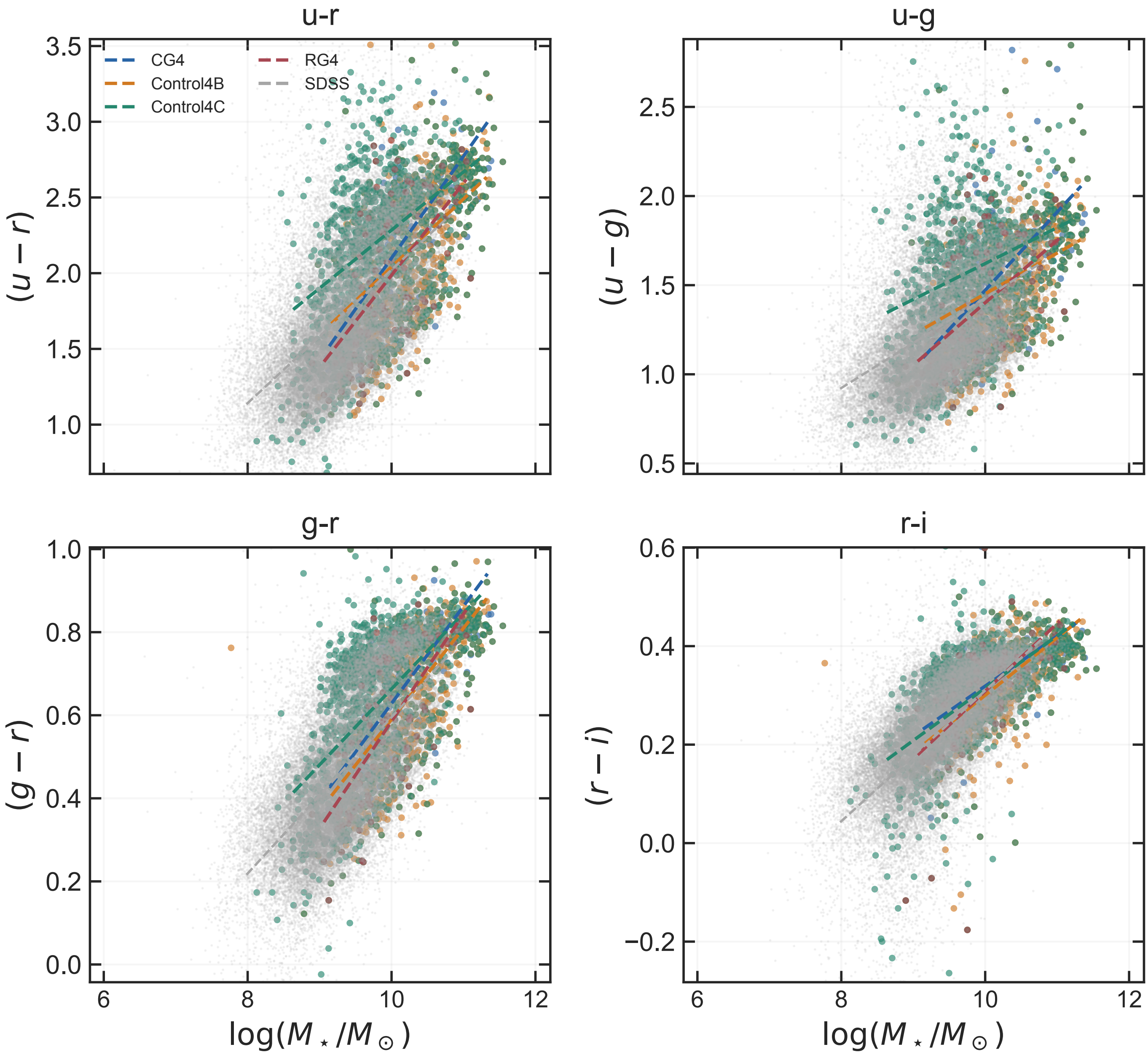

Fig. A.13. Optical colour as a function of stellar mass for CG_4, the three control catalogues, and the matched SDSS reference sample. Dashed lines show the fitted catalogue relations.

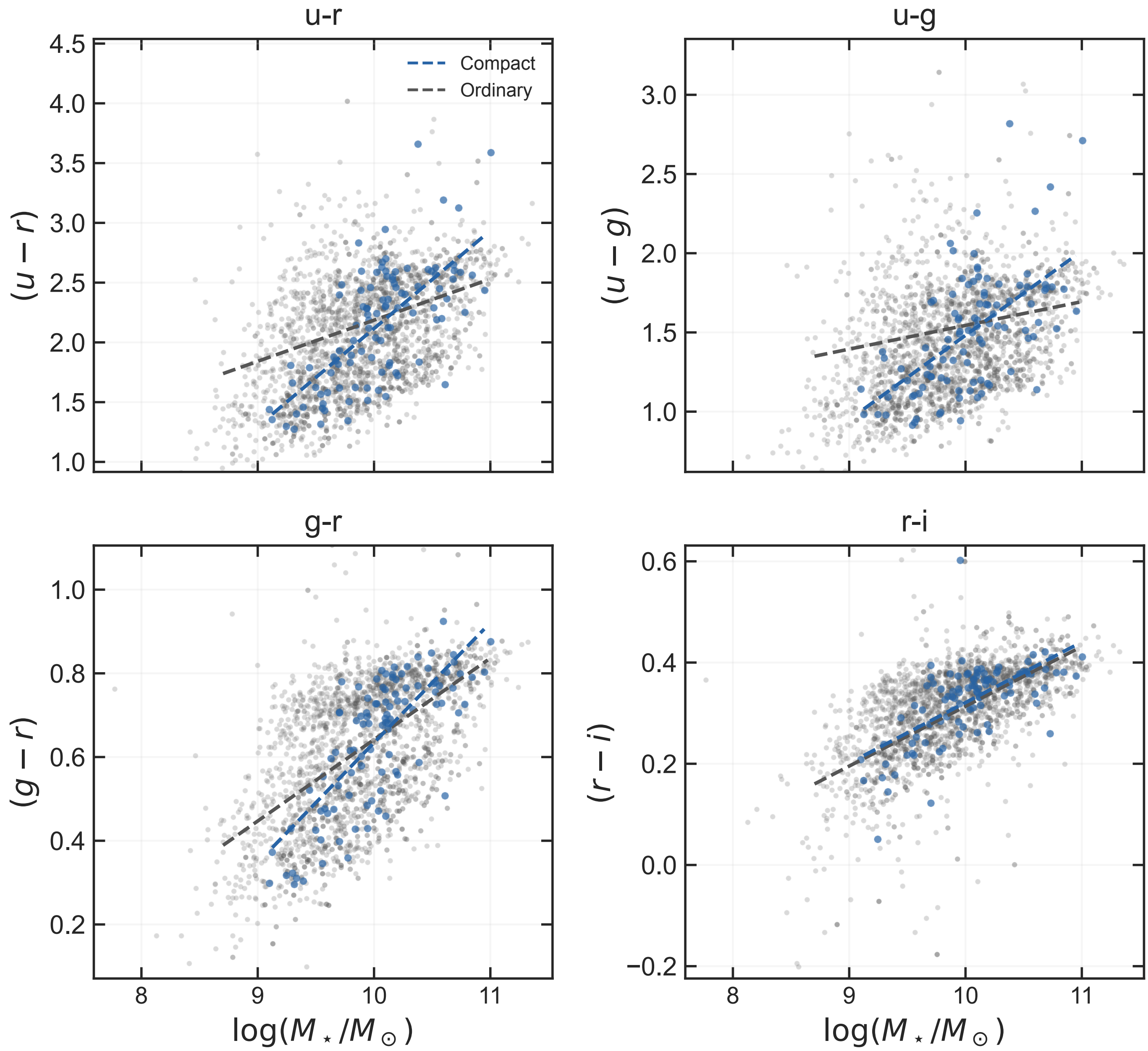

Fig. A.14. Satellite colour–mass relations in compact and ordinary groups. The fitted lines show both colour offsets at the reference mass and differences in slope.

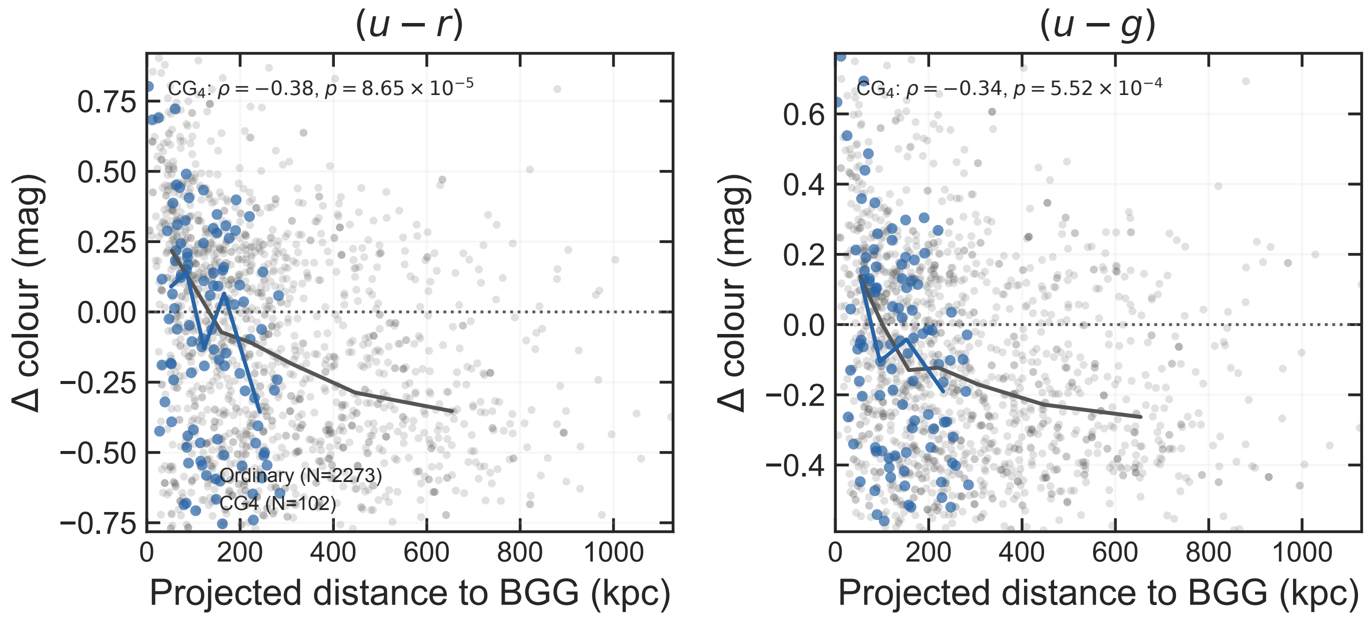

Fig. A.15. Mass- and redshift-adjusted satellite $(u-r)$ and $(u-g)$ residuals as a function of projected distance to the BGG. Lines connect binned medians for CG$_4$ and the pooled ordinary-group satellites; extreme plotting tails are omitted for readability, while the reported rank correlations use the full matched sample. More negative residuals are bluer than the ordinary-satellite reference relation.

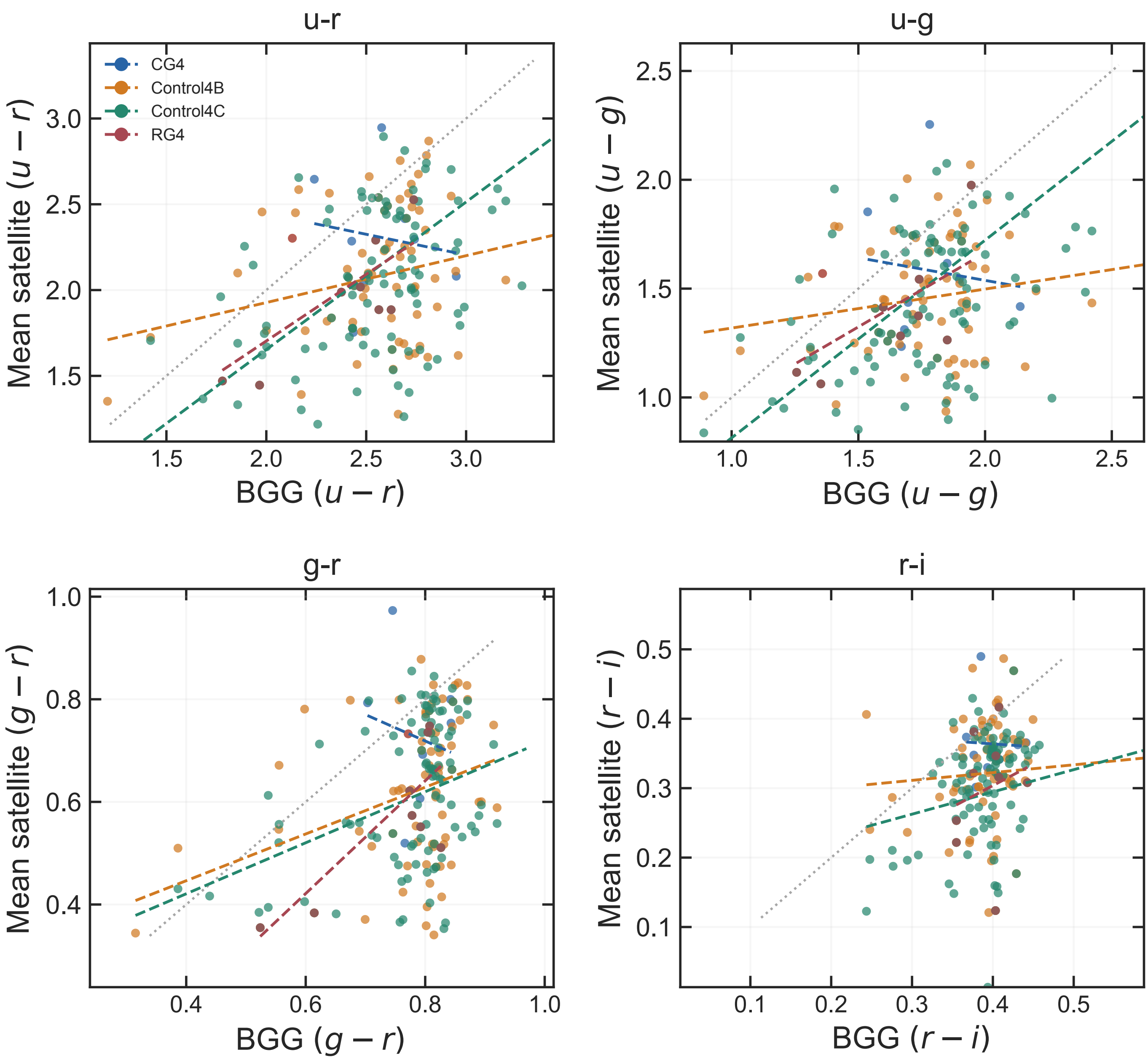

Fig. A.16. Mean satellite colour versus BGG colour for groups with matched photometry. The dotted line is the one-to-one relation and dashed lines are catalogue-specific fits.

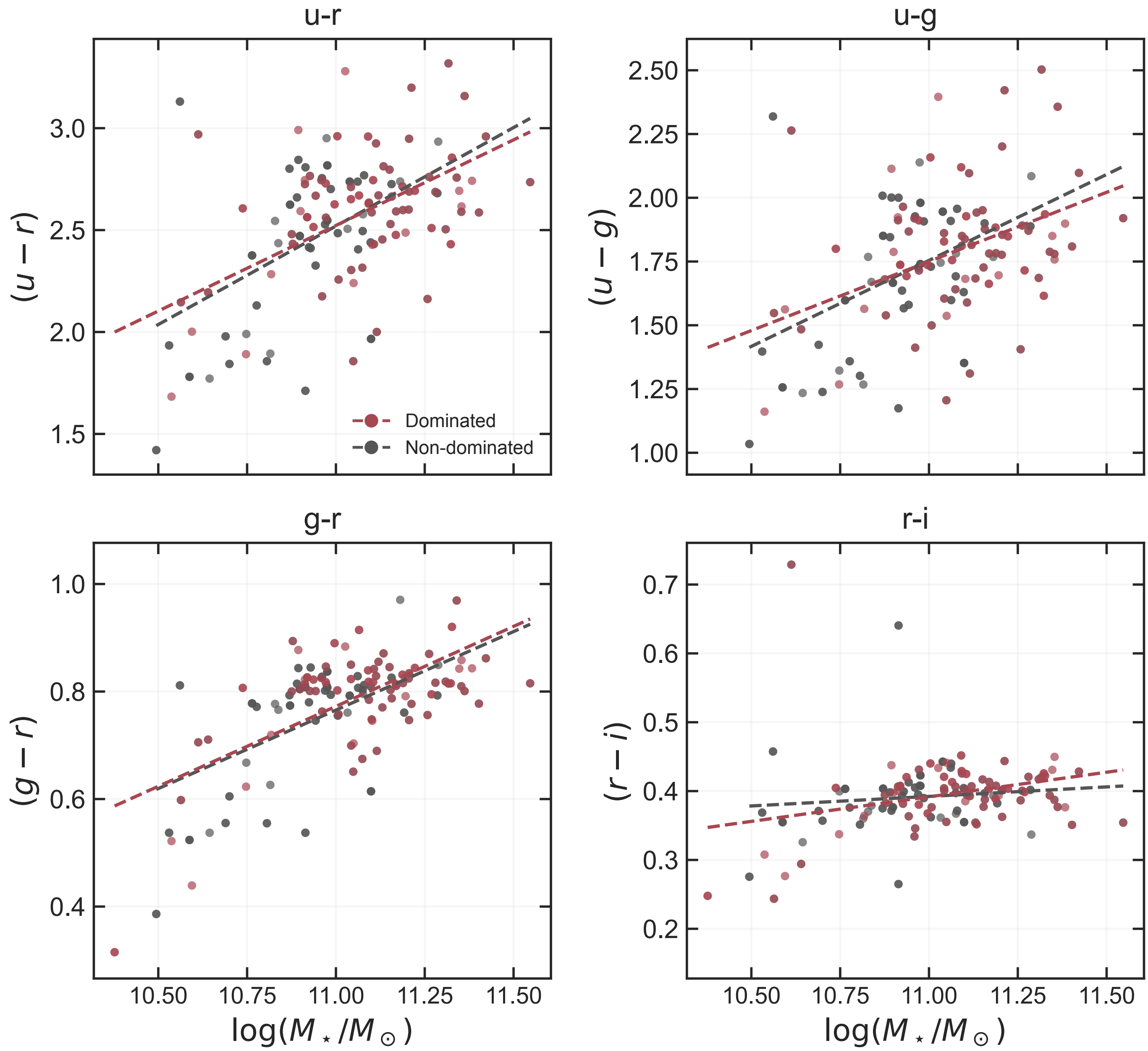

Fig. A.17. BGG colour–mass relations split by luminosity domination. Comparing the fitted relations at a common stellar mass separates colour effects from the different mass distributions.

A.4. Robustness and selection diagnostics

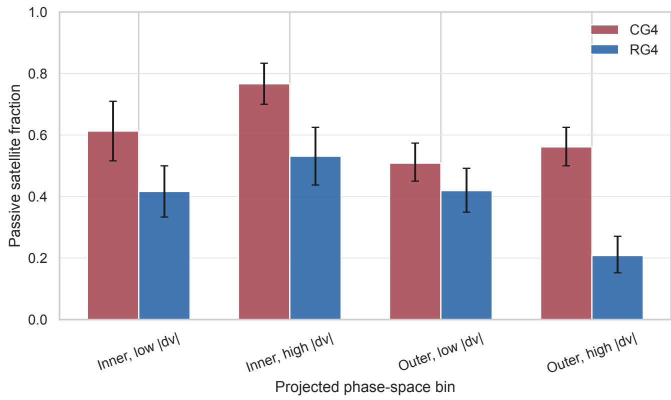

Fig. A.18. Passive satellite fraction in coarse projected phase-space bins. Inner and outer refer to projected-distance rank from the BGG; low and high velocity are split at $|\Delta v_{\mathrm{BGG}}|/\sigma_v = 1$.

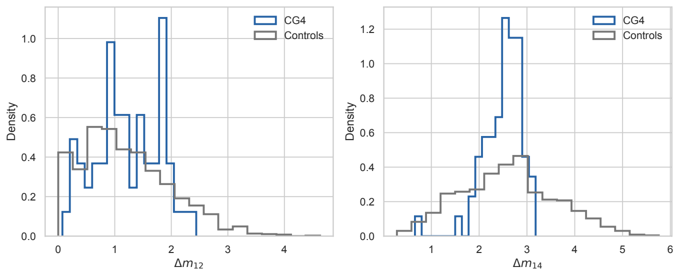

Fig. A.19. Distributions of the first–second and first–fourth magnitude gaps in CG_4 and the pooled controls.

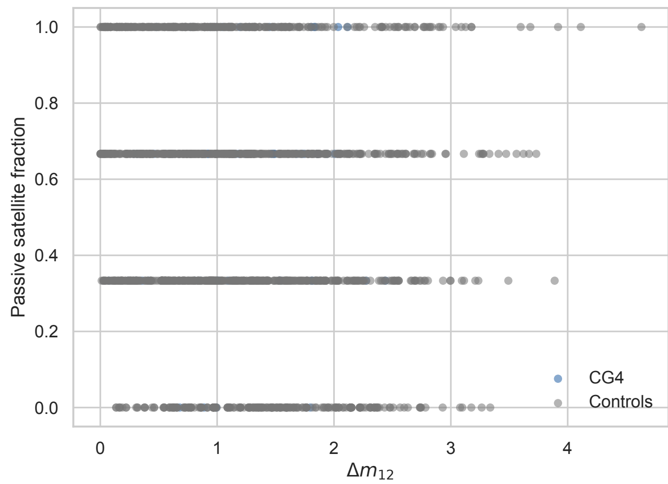

Fig. A.20. Passive satellite fraction as a function of the first–second magnitude gap.

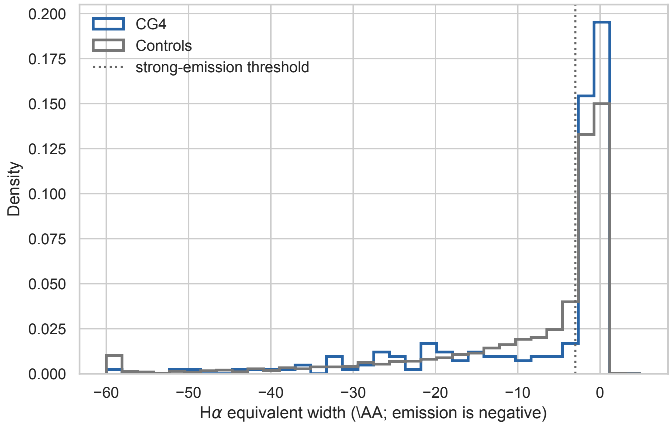

Fig. A.21. Hα equivalent-width distributions used for the limited recent-quenching diagnostic.

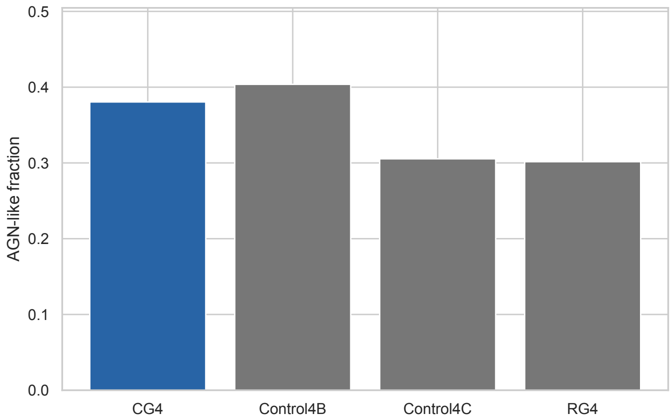

Fig. A.22. AGN-like fractions among galaxies with BPT-classifiable emission-line ratios. BPT-unclassified galaxies are not counted as non-AGN in this diagnostic.

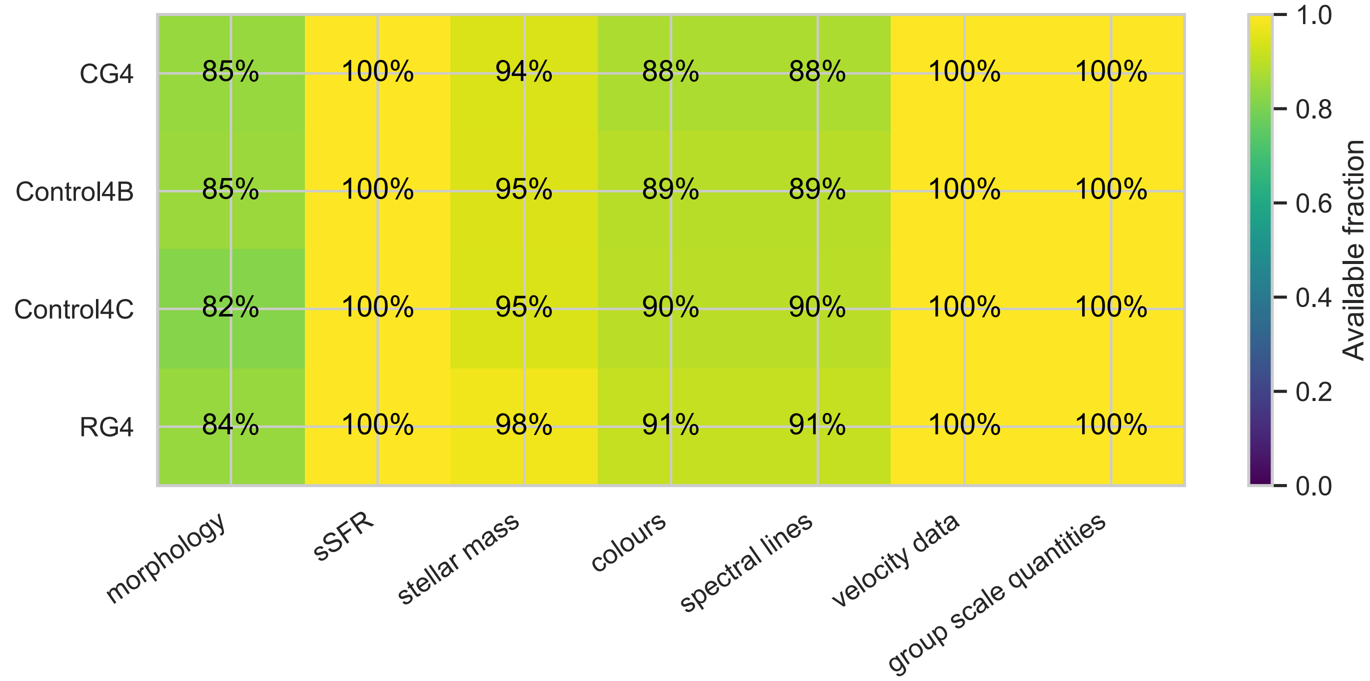

Fig. A.23. Availability fractions for the major derived quantities in each sample.

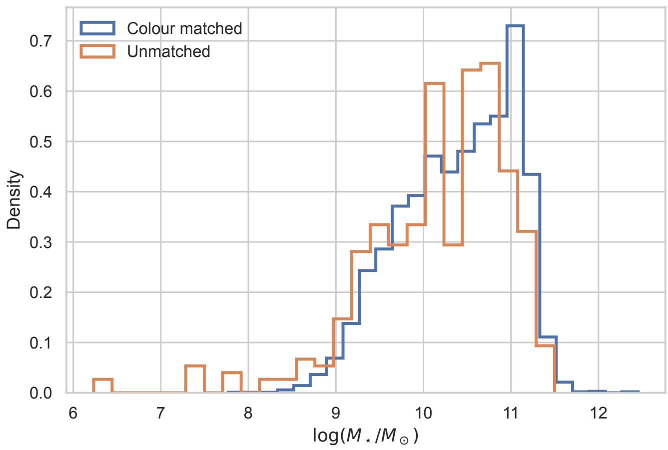

Fig. A.24. Stellar-mass distributions of galaxies with and without complete matched colour measurements.

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